Air Pollution and Extreme Heat Exposures for Young Children in Latin America and the Caribbean

Preliminary draft

The latest version of this paper is available at this link.

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Abstract

Air pollution and extreme heat pose important risks to human health and wellbeing, and children are particularly vulnerable to these hazards. We analyze how ambient exposures to these hazards are distributed across, and overlap with, the young child population of Latin America and the Caribbean. We then compare ambient exposure across socioeconomic and geographic groups. Using multiple satellite and administrative datasets available between 2000 and 2020, we analyze changes over time in the overlap between these groups and ambient hazards. Our findings highlight substantial increases in the mean and, in absolute terms, in the dispersion of ambient exposures from 2000 to 2020. Compared with 2000, in 2020 an additional 7.9 million children were exposed to air pollution above $10\,\mu g/m^3$, the World Health Organization limit for safe exposure until 2021. For heat stress, 3.5 million additional children were exposed to a Universal Thermal Climate Index (UTCI) above 32°C for at least 3 months of the year. Beyond increases in single-hazard ambient exposures, we find growing compound exposures. In 2020, 4.3 million children (9.2% of the child population) were exposed to both $PM_{2.5}$ above $15\,\mu g/m^3$ and at least 3 months of extreme heat, 3.2 million more than in 2000. Moreover, higher exposures overlap with more socioeconomically disadvantaged regions (low-income and low-Human Development Index), and we document differences associated with geospatial groups, such as remoteness, rurality, and Indigenous land status. These findings show that ambient exposure compounds socioeconomic disadvantage and threatens young children's health and wellbeing.

Keywords: Climate; Air Pollution; Extreme Heat; Young Children; Latin America and the Caribbean

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1 Introduction

The countries of Latin America and the Caribbean (LAC) face high and increasing exposure to air pollution and extreme heat (Rees et al. 2021; UNOCHA and UNDRR 2023; UNICEF 2026). Young children are especially vulnerable, particularly from conception to age five, when development is most sensitive. In this life-cycle stage, children are less able to regulate their body temperature during heat waves and are physiologically more sensitive to air pollution than adults (Rees et al. 2021; Landrigan et al. 2019). These risks are not equally distributed across the region and among children. LAC is among the most unequal regions in the world (UNECLAC 2023), with differences in income, access to health care, and public services. These inequalities can limit the ability to use adaptation strategies and shape both how much exposure children face and how households can adapt to these ambient exposures (Mullins and White 2020).

Although exposure to air pollution and extreme heat is increasingly well documented, evidence remains limited on how exposure has changed over time, and how it differs across socioeconomic and geographic groups in LAC. For instance, global assessments count children exposed to hazards at a point in time (UNICEF 2026), regional monitoring reports track extreme heat and air pollution separately (Hartinger et al. 2024), and studies of inequalities in exposure are largely confined to single countries or cities (Coelho and Requia 2025; Aguilar-Gomez, Cardenas, and Salas Diaz 2025; Chakraborti and Voorheis 2025). Global studies document the distribution of exposure to air pollution and extreme heat (Rentschler and Leonova 2023; Sager 2025; Tuholske et al. 2021), but do not focus on young children or compare socioeconomic and geographic groups within LAC. This paper uses gridded data on child populations, $PM_{2.5}$ pollution, and high temperature to study children aged 0 to 5 in LAC between 2000 and 2020. We build on the approach of Feng et al. (2024), who measure children's extreme heat exposure in China with child-population-weighted statistics, so that measures describe the exposure of the average child. Even though extreme heat and air pollution interactions can amplify health risks beginning in the prenatal period (Sun et al. 2020; Hu et al. 2024), combined exposures remain understudied in low- and middle-income countries (Raghavan et al. 2025). We investigate the overlap of these risks in LAC and how this has developed over time. We make three contributions to the literature. First, we study air pollution, extreme heat, and compound exposure, defined as children living in cells above both pollution and heat-severity thresholds. Second, we compare exposure across income, human development, remoteness, rurality, and Indigenous land. Third, we report beyond group means by documenting the dispersion of exposure within groups over time and by groups.

Our study combines child population estimates for 2000, 2010, and 2020 with gridded data on air pollution and extreme heat across 2,180 grid cells ($1° \times 1°$ resolution) covering countries and subregions of LAC (i.e., South America, Caribbean, Central America). Child population estimates, from 0 to 5 years old, come from WorldPop, aggregated from its 30 arc-second grid to the $1° \times 1°$ cell level. Cells with children contained 45 million children in 2000 and 46.9 million in 2020. Air pollution is measured by annual mean $PM_{2.5}$ concentrations from satellite-based estimates (Van Donkelaar et al. 2021), which we classify severity using the World Health Organization (WHO) guideline of $10\,\mu g/m^3$ and its interim targets (ITs) of 15 (IT3),

25 (IT2), and 35 $\mu g/m^3$ (IT1).[1]. Extreme heat is measured by the hourly Universal Thermal Climate Index (UTCI), a composite thermal-stress metric that integrates air temperature, radiation, wind, and humidity, from the ERA5-HEAT reanalysis (Napoli 2020), which we summarize as the share of the year above 32°C (a share of 0.25 corresponds to three months). We define compound exposure as a child living in a cell above both a $PM_{2.5}$ threshold and a heat duration threshold. Weighting each cell by its child population, we compute the mean exposure of the average child and the 20th and 80th percentiles of exposure. We compare these measures across income, human development, remoteness, rurality, and Indigenous land, with group assignments held fixed at the most recent year available. Because children are concentrated in a subset of cells, child-population weighting gives a more accurate picture of the exposure overlapping with the child population.

Our results show substantial increases in exposure between 2000 and 2020. Compared with 2000, about 8 million additional children in 2020 were exposed to $PM_{2.5}$ above $10\,\mu g/m^3$, the WHO's safe level until 2021, and 3.5 million additional children were exposed to a UTCI above 32°C for at least three months a year. We show that these increases reflect changes in the ambient exposure distribution, not changes in the children's population distribution. We also document increases in compound exposure. In 2020, 4.3 million children (9.2% of young children) lived where $PM_{2.5}$ exceeded $15\,\mu g/m^3$ and the UTCI exceeded 32°C for at least three months, 3.2 million more than in 2000. Moreover, these exposures are unequally distributed across socioeconomic and geographic groups. In 2020, the share of children facing compound exposure was 13% in the lowest income quartile, compared with 5% in the highest, and 16% in the lowest HDI quartile, compared with none in the highest. Among geographic groups, the share was 51% in the most remote quartile of cells, compared with 6% in the most accessible, and 40% in the most rural quartile, compared with 7% in the most urban. Children in Indigenous territories also faced higher compound exposure (13%) than other children (9%). These findings identify ambient exposure as an important dimension of vulnerability that compounds socioeconomic disadvantage and threatens the health and wellbeing of young children.

2 Data and Methods

2.1 Children and exposure data

We combine gridded data on children, air pollution, and heat on a common $1° \times 1°$ grid for 2000, 2010, and 2020. Child population is from WorldPop (Tatem 2017), which provides population by age at about 1 km resolution. To capture the young children population, we restrict it to ages 0–5. Air pollution is the annual mean $PM_{2.5}$ concentration from Donkelaar et al. (2016) at 0.01°, averaged within each cell. We classify exposure severity by using the World Health Organization (WHO) guideline and interim targets (IT4 to IT1), which correspond to 10, 15, 25, and 35 $\mu g/m^3$ (*WHO global air quality guidelines* 2021).

Extreme heat exposure is measured with the Universal Thermal Climate Index (UTCI) from the ERA5-HEAT dataset (Napoli 2020), which provides hourly data at 0.25°. UTCI is

1. More information on the WHO interim targets can be found in the WHO global air quality guidelines (*WHO global air quality guidelines* 2021).

an index of the human-perceived equivalent temperature that accounts for air temperature, humidity, wind, and radiant heat (Bröde et al. 2012). Following Feng et al. (2024), we compute for each cell the share of hours in a year above a UTCI threshold. Weighted by children, this is the Share of Time for the Average Child (STAC). As a severity benchmark, we express it in months per year. For example, a share of 0.25 corresponds to three months. Our main threshold is 32°C, the lower bound of strong heat stress (Bröde et al. 2012), we add results at 26°C and 38°C to our Supplementary Materials.

2.2 Socioeconomic and geographic groups

We assign each cell to groups using five layers and hold the categories fixed to the most recent year available for each layer, so that changes within a group do not reflect cells moving between groups. For socioeconomics layers, we use as income the GDP per capita in constant 2021 PPP dollars from the Becker Friedman Institute (Rossi-Hansberg and Zhang 2026), and human development is the Subnational Human Development Index from the Global Data Lab (Smits and Permanyer 2019). We divide cells into quartiles using 2020 values, assigning Q1 as the lowest chunks of the distribution. For geographical layers, accessibility is the travel time to the nearest city of at least 50,000 inhabitants in 2015 (Weiss et al. 2018), averaged within each cell and divided into quartiles (Q1 is the most remote). Land type separates cells whose centroid lies within an Indigenous territory in the LandMark database (LandMark 2026) from all other cells. Urbanity is defined by the share of each cell classified as urban in the Global Human Settlement Layer (Pesaresi et al. 2024). Because about 40% of cells contain no urban land, Q1 holds all cells with no urban land (the most rural) and Q2 to Q4 split the remaining cells into tertiles. The Supplementary Materials describe each layer in detail.

2.3 Statistics

To capture overlapping exposures, we merged cell-level data on air pollution, heat, and children, and count of children exposed to both poor air quality and months of extreme heat. For example, $PM_{2.5}$ above IT3 ($15\,\mu g/m^3$) and at least three months above 32°C, for the region, its subregions, countries, and groups. For each group and year, we report the child-weighted mean and the 20th and 80th percentiles of exposure, and the difference between the two percentiles (P80−P20) as a measure of dispersion. To separate environmental change from changes in where children live, we recompute each statistic holding the 2000 distribution of children fixed. All statistics are child-population-weighted, and their formal definitions are in the Supplementary Materials.

3 Results

3.1 Ambient exposures

Our results show that between 2000 and 2020, the number of children in the region facing lower air quality and rising extreme heat exposure increased sharply. Our distributional results are in Figure 1, which presents the distributions of $PM_{2.5}$ and shares of hours of the year above 32°C for children in years 2000, 2010, and 2020, using cell-level statistics. These distributions are computed using child population, pollution, and temperature data aggregated at the cell

level. To have benchmarks for severity of exposure, we add vertical lines to indicate WHO's Interim Targets (ITs) and the share of hours of exposure summarized in months thresholds. We complement the visualization with shaded areas to make clear the range of values associated with each severity category. Figure 2 reports the corresponding number of children (in millions) in each severity category for $PM_{2.5}$ (panel a), months above 32°C (panel b), and both hazards combined (panel c).

Air pollution. The mean child air pollution exposure in the region increased from 13.05 $\mu g/m^3$ in 2000 to 15.96 $\mu g/m^3$ in 2020 (Table 1).[2] Figure 2a shows the shift across severity categories. In 2020, 40.6 million children (87% of the child population) were exposed to air quality that did not meet the WHO limit of 10 $\mu g/m^3$, an increase of 7.9 million relative to 2000. Children also moved up the severity scale. The number in the 15–25 $\mu g/m^3$ category (IT3) rose from 11.2 to 20.7 million, and the number above 25 $\mu g/m^3$ (IT2 and above) rose from 0.4 to 3.5 million. In 2020, 0.2 million children were exposed to levels above 35 $\mu g/m^3$ (IT1), the most severe category, which had no child population in 2000.

Subregional trajectories differ markedly. Mean $PM_{2.5}$ rose most in South America, from 11.5 to 15.2 $\mu g/m^3$, and by about 1.2 $\mu g/m^3$ in the Caribbean and Central America between 2000 and 2020. Central America remains the most polluted subregion (18.1 $\mu g/m^3$ in 2020) and is the only one with children above the highest threshold, 35 $\mu g/m^3$. Moreover, 16% of its children live above 25 $\mu g/m^3$. In the Caribbean, all children live below 15 $\mu g/m^3$ in 2000, but 38% are above that level in 2020.

Heat stress. Focusing on extreme heat, children were more exposed in 2020 than in 2000. The average child spent about 9.6% of the year (around one month) above a UTCI of 32°C in 2000, rising to 12.5% in 2020. Figure 2b shows how children are distributed across months of exposure. The number of children exposed to at least three months a year above 32°C nearly doubled, from 4.0 to 7.5 million, an increase of 3.5 million that represents 16% of the child population in 2020. Exposure to the longest durations grew fastest. The number of children exposed to four to five months rose from 0.5 to 2.5 million, and none were exposed to five or more months.

The Caribbean has the largest increase in the share of the year above 32°C, from 14% to 18%, about 15 additional days, followed by South America and Central America. Long-duration exposure grew fastest in South America, where the number of children facing four to five months a year above 32°C rose from 0.47 to 2.32 million (7.6% of its children in 2020). In Central America, the share facing three to four months rose from 4% to 10%.

Joint exposure to air pollution and heat stress. Overlapping exposure to air pollution and heat rose sharply (Figure 2c). In 2020, 4.3 million children (9.2%) lived where $PM_{2.5}$ exceeded 15 $\mu g/m^3$ (IT3) and temperatures exceeded 32°C for at least three months a year, up from 1.1 million (2.4%) in 2000. Exposure to both IT3 pollution and at least four months of heat rose from 0.1 to 1.7 million children (3.6% in 2020). At the higher pollution threshold of 25 $\mu g/m^3$ (IT2), no children were exposed to both hazards in 2000, but in 2020 0.8 million faced at least

2. We report mean statistics in Table C.1, and shares of children above thresholds in Tables C.2, C.3, C.4 and C.5.

three months of heat and 0.6 million at least four months. Compound exposure is concentrated in South America and, increasingly, Central America, and is absent in the Caribbean (Table 2). In South America, 10.4% of children faced IT3 pollution and at least three months of heat in 2020 (2.2% in 2000), and 4.9% faced IT3 pollution and at least four months (0.2% in 2000). In Central America, the corresponding shares were 7.8% (3.1% in 2000) and 1.2% (none in 2000).

Population shifts. The number of children exposed can rise either because ambient exposure worsens or because of changes in children's location. To test whether population changes drive our results, we weight the 2010 and 2020 cell-level distributions of $PM_{2.5}$ and of the share of the year above 32°C by the 2000 spatial distribution of children. In this counterfactual, ambient conditions change over time, but the population distribution is fixed. The counterfactual distributions are nearly identical to the observed ones (Figure 1, panels b and d), indicating that environmental change, not population redistribution, drives the increases in both mean exposure and its dispersion.

3.2 Exposure by socioeconomic and geographical groups

In addition to overlapping air pollution and extreme heat distributions with the child population, we add socioeconomic and geospatial information layers to investigate if ambient exposures overlap with different groups. Figure 3 shows the mean, 20th percentile, and 80th percentile of exposure for each of these groups. Table 1 reports these statistics together with measures of inequality, and Table 2 shows the share of children in each of these groups who are exposed to both extreme heat and high levels of air pollution.

Socioeconomic layer To characterize socioeconomic groups, we use cell-level estimates of GDP per capita and the subnational Human Development Index (HDI). Both are important stratifiers because they are strongly correlated with the capacity to mitigate the effects of ambient exposures on children and with other vulnerabilities, such as the availability and quality of public services. To rank groups, we split the distributions of income and HDI into quartiles, where Q1 denotes the lowest values.

The results in Figure 3 show that children in poorer and less developed areas face both the highest ambient exposure and a wider dispersion of exposure than children in richer and more developed areas. For air pollution, mean $PM_{2.5}$ exposure in the lowest income quartile rose from 13.7 to 17.9 $\mu g/m^3$ between 2000 and 2020. Over the same period, the range between the 20th and 80th percentiles widened from 11.0 to 15.5 $\mu g/m^3$. In the highest income quartile, the mean rose only from 11.4 to 12.8 $\mu g/m^3$, and the P20–P80 range stayed narrow (3.5 to 4.0 $\mu g/m^3$). As a result, the gap in mean exposure between the bottom and top income quartiles doubled, from 2.4 to 5.0 $\mu g/m^3$. The lowest HDI quartile shows the highest mean exposure of any group, at 19.3 $\mu g/m^3$ in 2020, up from 15.5, about 4 $\mu g/m^3$ above WHO Interim Target 3. Its dispersion remained roughly stable over the period.

For extreme heat, the lowest income and HDI quartiles both went from about 1.5 to 1.9 months per year above 32°C (shares of 0.125 to 0.161 and 0.126 to 0.162). Their P20–P80 ranges widened from about 2.4 to 3.1 months for income and from 2.7 to 3.2 months for HDI. By contrast, children in the highest HDI quartile spent less than one month above 32°C in 2020 (0.7 months), and the range of their exposure stayed below one month.

Figure 4 makes this pattern even clearer. It shows the share of children exposed to both air pollution and extreme heat in the lowest and highest income and HDI quartiles. In both dimensions, exposure rises most among children in the lowest quartile. In the lowest income quartile, the share of children exposed to $PM_{2.5}$ above IT3 (15 $\mu g/m^3$) and at least three months above 32°C rose by 8.7 percentage points between 2000 and 2020. With a four-month heat threshold, it rose by 4.8 points. In the highest income quartile, the corresponding increases were 4.1 and 1.3 points. The contrast is starker for HDI. In the highest HDI quartile, no children face this combined exposure in any year. In the lowest quartile, the share rose by 12.3 points under the three-month threshold and by 4.6 points under the four-month threshold.

Geographical groups For the geographic layers, we stratify cells along three dimensions. First, we classify cells by land tenure into two groups, indigenous and non-indigenous land. Second, we group cells into remoteness quartiles based on travel time to the nearest city of at least 50,000 inhabitants (Weiss et al. 2018). Q1 contains the most remote cells and Q4 the most accessible. Third, we group cells by urbanization, measured as the share of urban land cover within each $1° \times 1°$ cell in 2020. Because about 40% of cells contain no urban land, the quartile cut-offs are not unique. We assign all cells with zero urban share to Q1 and split the remaining cells into Q2–Q4 by tertiles of urban share. Q1 thus captures entirely rural cells and Q4 the most urbanized.

4 Discussion

Exposure to air pollution and extreme heat in early childhood can have lasting consequences for health and development, so knowing where these hazards overlap with young children is a central concern. The capacity to mitigate these effects is also strongly linked to socioeconomic and geographic characteristics. In this study, we combine data on the child population, ambient exposures, and socioeconomic and geographic characteristics to examine how these layers overlap in Latin America and the Caribbean, a region that is unequal in many dimensions.

We show that between 2000 and 2020, exposure to air pollution and extreme heat increased, and the gap between the most and least exposed children (P80 minus P20) widened in absolute terms. The number of children facing both $PM_{2.5}$ above $15\,\mu g/m^3$ and at least three months of extreme heat rose from 1.1 to 4.3 million. Children in the poorest and least developed areas bear the highest burden of both pollution and heat. We also find that geography shapes exposure: children in remote and rural areas face the most heat and the largest increases in air pollution.

Our results contribute to the environmental justice literature in four ways. First, we study a population and region that have received less attention: young children in Latin America and the Caribbean. The existing evidence on environmental inequality in the region focuses largely on cities (Fernández, Koplow-Villavicencio, and Montoya-Tangarife 2023), whereas we cover the whole region and a range of groups, including rural and remote areas, where we find the highest burdens, and Indigenous lands, where compound exposure is higher. Second, we consider cumulative burdens. Rather than studying one hazard at a time, we analyze air pollution and heat jointly, in line with recent calls to account for the number and intensity of

overlapping burdens (Mohai and Lee 2026). Third, we weight every exposure measure by the spatial distribution of young children, so our statistics describe the conditions the average child experiences, not the average location or the average resident. Studies that report area-weighted or national averages can misstate children's exposure where children are concentrated in particularly polluted or hot places. We also report full distributions rather than only means, which reveals inequality within groups as well as between them (Feng et al. 2024; Santos et al. 2024). Fourth, we track these overlaps over two decades. Our findings are consistent with global and regional evidence that environmental burdens fall disproportionately on more vulnerable populations (Rees et al. 2021; Santos et al. 2024; Drupp et al. 2025). Beyond documenting these gaps, we show that they widened between 2000 and 2020, and that the rise in exposure reflects changes in the environment rather than shifts in where children live.

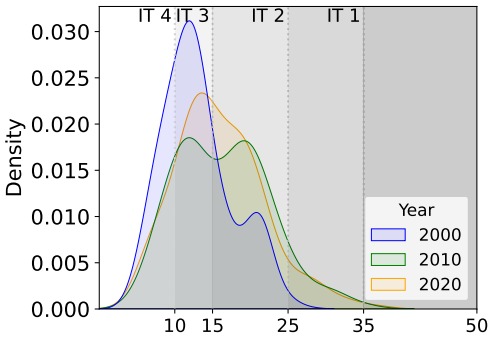
Our data sources and construction have several limitations. First, aggregating to $1^\circ \times 1^\circ$ cells (about 110 km at the equator) averages over substantial variation within each cell, such as differences between urban and rural areas or within cities. It can also introduce boundary errors, especially for small countries and islands. Second, our exposure and population measures come from satellite retrievals, reanalysis models, and gridded population estimates, which rely on interpolation for missing periods or locations. Measurement error from these sources carries into our estimates. To minimize this concern, we use published and validated datasets from the scientific literature. Third, because of data availability, we hold the geographic and socioeconomic groups fixed at their 2020 values (2015 for accessibility). Some characteristics, such as land type and remoteness, change little over two decades, and relative rankings by income and development tend to be persistent, but some cells may have moved between quartiles over the period. Fourth, our measures capture ambient outdoor conditions. They do not reflect indoor or household exposure, time spent outdoors, or access to cooling, so actual personal exposure may differ from ambient levels. In addition, the 32°C UTCI threshold was developed for adults, and young children may experience heat stress at lower levels. Finally, we document exposure, not its consequences: linking these exposure patterns to child health and development outcomes is an important next step.

Our study draws attention to the growing environmental pressures faced by young children in Latin America and the Caribbean. These pressures fall most heavily on the children with the least socioeconomic capacity to cope with them: those in poorer, less developed, and more remote areas. This has direct implications for policy. Social safety nets, early childhood services, and health systems should account for environmental exposure when identifying the populations most at risk, for example by prioritizing areas where pollution and heat overlap, extending heat-health warning systems to rural and remote areas, and adapting childcare and school schedules during extreme heat. Without such action, widening exposure gaps may translate into widening gaps in health, development, and long-term outcomes between children in poorer and in more prosperous regions.

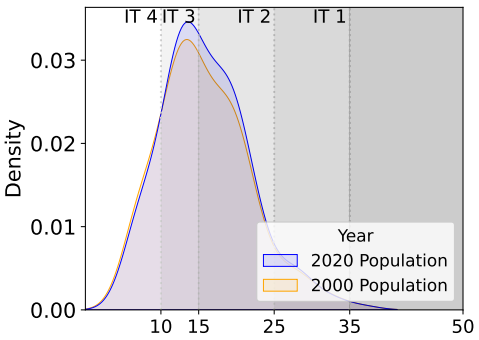
Figures and Tables

Figure 1: The distribution of air pollution exposure and heat exposure for children

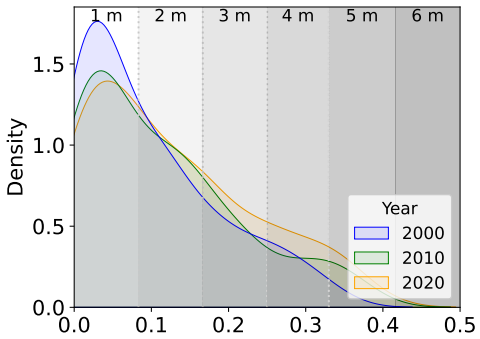
(a) $PM_{2.5}$, population of each year



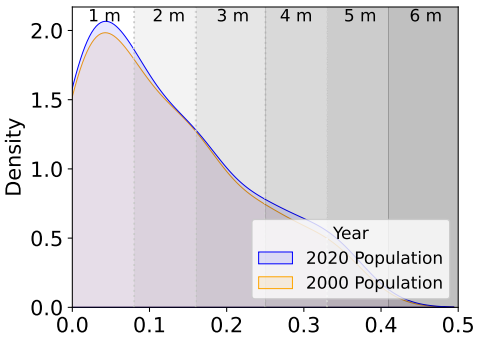
(b) $PM_{2.5}$ in 2020; 2020 vs. 2000 population weights



(c) Share of year above 32°C, population of each year



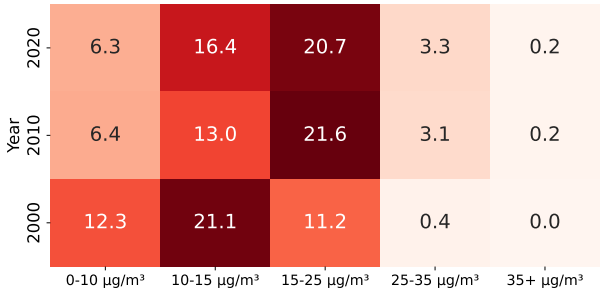
(d) Share of year above 32°C in 2020; 2020 vs. 2000 population weights



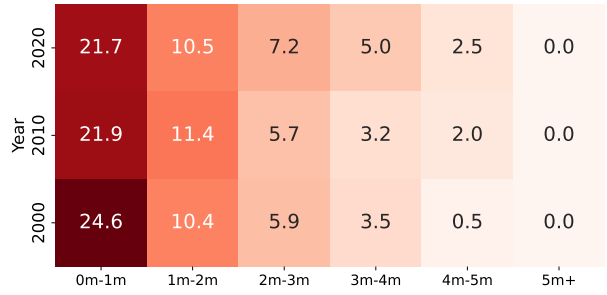
Notes: Kernel density distributions of cell-level $PM_{2.5}$ levels (top row) and share of the year above 32°C (bottom row), weighted by cell-level child population. The left panels show 2000, 2010, and 2020, each weighted by the child population of that year. The right panels show the 2020 distribution weighted by the 2020 child population (blue) and by the 2000 child population (orange), which separates changes in environmental exposure from changes in where children live. Background shading indicates severity ranges; vertical dotted lines mark the WHO interim targets (IT) for $PM_{2.5}$ and one-month steps for heat (m = months above 32°C).

Figure 2: Children (in millions) exposed to air pollution and heat thresholds in LAC

(a) $PM_{2.5}$ Interim Targets



(b) Months above 32°C

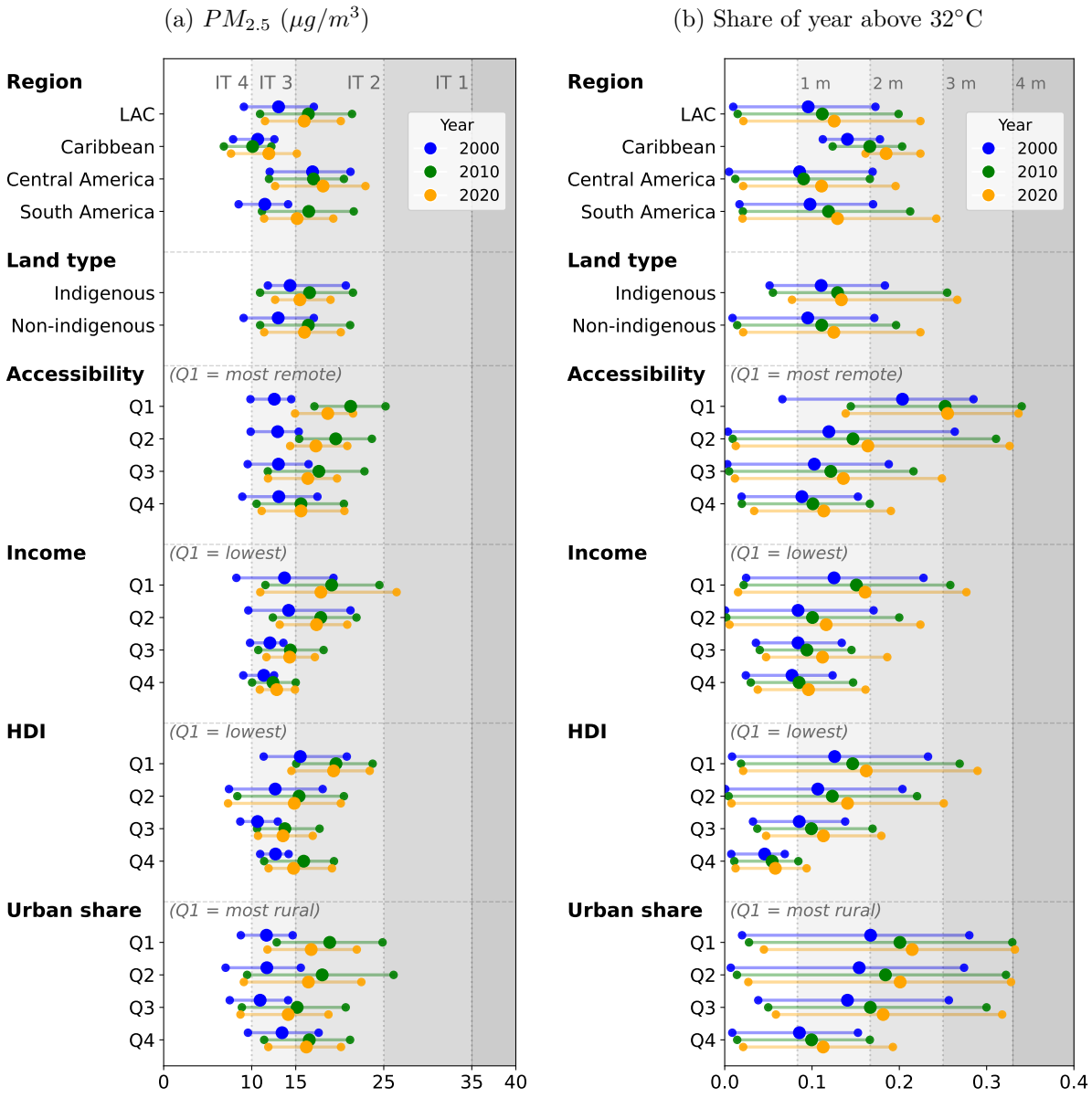


(c) Combined thresholds



Notes: Number of children (in millions) exposed to each threshold in 2000, 2010, and 2020. Panel (a) classifies children by $PM_{2.5}$ concentration ($\mu g/m^3$) following the WHO interim targets: guideline (< 10), IT4 (10–15), IT3 (15–25), IT2 (25–35), and IT1 (35+). Panel (b) classifies children by the number of months per year above 32°C. Panel (c) shows children exposed to both $PM_{2.5}$ above 15, 25, or 35 $\mu g/m^3$ and at least 3 or 4 months above 32°C. Values are obtained by overlapping cell-level $PM_{2.5}$, UTCI, and child population data.

Figure 3: Group-specific distributional ranges across all years



Notes: The large dot is the child-population-weighted mean and the small dots mark the 20th and 80th percentiles (joined by a line) of cell-level $PM_{2.5}$ levels (left) and share of the year above 32°C (right) in each group. Within each row, the dots from top to bottom correspond to 2000, 2010, and 2020. Groups are region (all of Latin America and the Caribbean pooled, and its Caribbean, Central America, and South America subregions), land type (indigenous vs. non-indigenous territories) and quartiles of accessibility (travel time to the nearest city; Q1 most remote, Q4 most accessible), income (GDP per capita at PPP, 2021 constant prices; Q1 lowest), HDI (Q1 lowest), and urban share (Q1 most rural). Group assignments use 2020 values (2015 for accessibility) and are held constant across years. Background shading indicates threshold ranges; vertical dotted lines mark the WHO interim targets (IT) for $PM_{2.5}$ and months above 32°C (m).

Figure 4: Overlapping air pollution and heat exposure by income group



Notes: Share (in percent) of children within each group exposed to both $PM_{2.5}$ and heat above the stated thresholds in 2000, 2010, and 2020 (m = months above 32°C). Columns combine a $PM_{2.5}$ threshold (> 15, > 25, or > 35 $\mu g/m^3$) with a heat threshold (> 3 or > 4 months). The top row shows the lowest (Q1) and highest (Q4) income quartiles, defined by cell-level GDP per capita at PPP (2021 constant prices) in 2020; the bottom row shows the lowest (Q1) and highest (Q4) HDI quartiles, defined by 2020 subnational HDI. Group assignments are held constant across years.

Tables

Table 1: Child exposure to PM 2.5 and heat: levels and inequality by group and year

		PM 2.5 ($\mu g/m^3$)				Share of year above 32°C			
Group	Year	Mean	P20	P80	P80−P20	Mean	P20	P80	P80−P20
Region									
LAC	2000	13.05	9.11	17.06	7.95	0.096	0.010	0.173	0.163
	2010	16.44	10.95	21.39	10.44	0.112	0.015	0.199	0.184
	2020	15.96	11.51	20.12	8.61	0.125	0.021	0.224	0.203
Caribbean	2000	10.69	7.89	12.57	4.69	0.141	0.112	0.178	0.065
	2010	10.10	6.83	12.23	5.40	0.166	0.124	0.203	0.080
	2020	11.94	7.65	15.12	7.48	0.185	0.161	0.224	0.063
Central America	2000	16.91	12.05	21.22	9.17	0.086	0.005	0.169	0.165
	2010	17.01	11.96	20.47	8.51	0.090	0.012	0.166	0.154
	2020	18.10	12.66	22.92	10.26	0.111	0.021	0.196	0.175
South America	2000	11.49	8.51	14.13	5.62	0.098	0.017	0.170	0.153
	2010	16.48	11.16	21.59	10.42	0.119	0.021	0.212	0.192
	2020	15.16	11.41	19.27	7.86	0.129	0.020	0.242	0.222
Land type									
Indigenous	2000	14.36	11.84	20.69	8.86	0.110	0.051	0.184	0.132
	2010	16.57	10.95	21.50	10.55	0.129	0.055	0.255	0.200
	2020	15.47	12.66	18.95	6.29	0.134	0.077	0.266	0.189
Non-indigenous	2000	13.01	9.08	17.06	7.98	0.095	0.009	0.171	0.162
	2010	16.43	10.96	21.18	10.23	0.111	0.014	0.196	0.182
	2020	15.99	11.42	20.12	8.70	0.125	0.021	0.224	0.203
Accessibility *(Q1 = most remote)*									
Q1	2000	12.57	9.86	14.49	4.63	0.204	0.066	0.285	0.219
	2010	21.24	17.12	25.20	8.08	0.252	0.144	0.340	0.196
	2020	18.64	14.92	21.51	6.59	0.255	0.139	0.337	0.198
Q2	2000	12.95	9.88	15.34	5.46	0.119	0.004	0.263	0.260
	2010	19.54	15.39	23.65	8.26	0.147	0.009	0.311	0.302
	2020	17.29	14.37	20.86	6.49	0.164	0.013	0.326	0.314
Q3	2000	13.04	9.54	16.46	6.91	0.103	0.003	0.188	0.185
	2010	17.63	11.83	22.81	10.98	0.121	0.005	0.216	0.211
	2020	16.37	11.85	19.69	7.84	0.136	0.012	0.249	0.237
Q4	2000	13.08	8.93	17.46	8.53	0.088	0.019	0.153	0.133
	2010	15.60	10.55	20.47	9.92	0.101	0.020	0.166	0.147
	2020	15.59	11.13	20.53	9.40	0.113	0.034	0.190	0.157
Income *(Q1 = lowest)*									
Q1	2000	13.72	8.26	19.29	11.03	0.125	0.025	0.228	0.203
	2010	19.08	11.55	24.50	12.95	0.151	0.022	0.258	0.237
	2020	17.86	10.97	26.46	15.48	0.161	0.015	0.277	0.261
Q2	2000	14.19	9.61	21.22	11.61	0.084	0.000	0.171	0.170
	2010	17.85	12.40	21.90	9.50	0.101	0.002	0.200	0.198
	2020	17.36	13.18	20.85	7.67	0.116	0.005	0.224	0.219
Q3	2000	12.07	9.81	13.60	3.79	0.084	0.036	0.134	0.098
	2010	14.39	10.75	18.17	7.42	0.094	0.040	0.145	0.105
	2020	14.31	11.67	17.17	5.50	0.112	0.047	0.186	0.139
Q4	2000	11.37	9.04	12.54	3.50	0.077	0.024	0.124	0.099

Table 1: Child exposure to PM 2.5 and heat: levels and inequality by group and year

		PM 2.5 ($\mu g/m^3$)				Share of year above 32°C			
Group	Year	Mean	P20	P80	P80−P20	Mean	P20	P80	P80−P20
	2010	12.42	10.04	15.00	4.96	0.085	0.030	0.147	0.117
	2020	12.84	10.92	14.91	3.99	0.096	0.038	0.161	0.123
HDI *(Q1 = lowest)*									
Q1	2000	15.52	11.36	20.80	9.44	0.126	0.008	0.233	0.224
	2010	19.57	15.05	23.74	8.68	0.147	0.019	0.269	0.250
	2020	19.30	14.51	23.40	8.90	0.162	0.021	0.290	0.268
Q2	2000	12.67	7.42	18.06	10.64	0.107	0.001	0.204	0.203
	2010	15.38	8.37	20.47	12.10	0.123	0.004	0.220	0.216
	2020	14.83	7.32	20.12	12.80	0.141	0.008	0.251	0.243
Q3	2000	10.66	8.70	12.96	4.26	0.085	0.032	0.138	0.106
	2010	13.77	10.61	17.71	7.10	0.099	0.037	0.169	0.132
	2020	13.55	10.72	16.93	6.21	0.113	0.047	0.179	0.132
Q4	2000	12.70	10.96	14.19	3.23	0.046	0.007	0.069	0.061
	2010	15.90	11.41	19.34	7.93	0.054	0.011	0.084	0.073
	2020	14.75	11.92	19.12	7.20	0.058	0.012	0.094	0.081
Urban share *(Q1 = most rural)*									
Q1	2000	11.67	8.76	14.65	5.89	0.167	0.020	0.280	0.260
	2010	18.86	12.84	24.86	12.02	0.201	0.028	0.329	0.302
	2020	16.76	11.79	21.94	10.15	0.215	0.045	0.332	0.288
Q2	2000	11.72	7.03	15.58	8.54	0.154	0.007	0.274	0.267
	2010	18.01	9.48	26.12	16.64	0.184	0.014	0.322	0.308
	2020	16.44	9.11	22.45	13.35	0.201	0.027	0.328	0.301
Q3	2000	10.96	7.50	14.13	6.63	0.141	0.038	0.257	0.218
	2010	15.16	8.88	20.68	11.79	0.167	0.050	0.300	0.250
	2020	14.14	8.72	18.72	10.00	0.181	0.059	0.318	0.259
Q4	2000	13.45	9.57	17.60	8.03	0.086	0.009	0.153	0.144
	2010	16.54	11.41	21.17	9.76	0.099	0.014	0.166	0.152
	2020	16.21	11.88	20.14	8.26	0.113	0.021	0.193	0.171

Note: Child-population-weighted mean, 20th percentile (P20), and 80th percentile (P80) of cell-level $PM_{2.5}$ ($\mu g/m^3$) and of the share of the year above 32°C; the group statistics are the ones plotted in Figure 3. Dispersion of exposure within a group is measured by the difference P80−P20, in $\mu g/m^3$ for $PM_{2.5}$ and in share of the year for heat. LAC is Latin America and the Caribbean. Land type distinguishes indigenous and non-indigenous territories. Quartiles refer to accessibility (Q1 most remote), income (GDP per capita, Q1 lowest), HDI (Q1 lowest), and urban share (Q1 most rural); group assignments use 2020 values and are held constant across years.

Table 2: Share of children (%) exposed to compound air pollution and heat by group and year

Group	Year	$> 15\ \mu g/m^3$ & > 3 months	$> 15\ \mu g/m^3$ & > 4 months	$> 25\ \mu g/m^3$ & > 3 months	$> 25\ \mu g/m^3$ & > 4 months
Region					
LAC	2000	2.40	0.13	0.00	0.00
	2010	7.48	2.86	1.81	0.79
	2020	9.22	3.55	1.66	1.19
Caribbean	2000	0.00	0.00	0.00	0.00
	2010	0.00	0.00	0.00	0.00
	2020	0.00	0.00	0.00	0.00
Central America	2000	3.11	0.00	0.00	0.00
	2010	1.55	0.00	1.32	0.00
	2020	7.78	1.19	2.04	1.19
South America	2000	2.21	0.20	0.00	0.00
	2010	10.54	4.29	2.12	1.19
	2020	10.40	4.86	1.57	1.25
Land type					
Indigenous	2000	3.06	0.00	0.00	0.00
	2010	15.21	6.11	4.34	2.59
	2020	12.62	5.76	3.26	2.41
Non-indigenous	2000	2.38	0.14	0.00	0.00
	2010	7.19	2.73	1.71	0.72
	2020	9.03	3.42	1.58	1.12
Accessibility *(Q1 = most remote)*					
Q1	2000	15.16	0.00	0.00	0.00
	2010	59.32	34.41	18.25	13.53
	2020	51.15	25.82	11.61	10.96
Q2	2000	13.34	1.95	0.00	0.00
	2010	26.27	14.31	9.11	4.86
	2020	29.64	16.55	7.88	5.73
Q3	2000	2.33	0.00	0.00	0.00
	2010	9.03	3.20	1.07	0.78
	2020	9.22	3.18	0.74	0.59
Q4	2000	1.01	0.00	0.00	0.00
	2010	3.53	0.64	0.80	0.00
	2020	5.54	1.47	0.93	0.55
Income *(Q1 = lowest)*					
Q1	2000	4.03	0.00	0.00	0.00
	2010	12.44	3.19	2.69	0.43
	2020	12.67	4.77	2.96	1.65
Q2	2000	2.33	0.46	0.00	0.00
	2010	8.85	6.03	2.51	1.98
	2020	10.53	5.94	2.28	2.22
Q3	2000	1.37	0.00	0.00	0.00
	2010	3.52	0.44	0.69	0.22
	2020	6.38	0.82	0.29	0.11
Q4	2000	1.23	0.00	0.00	0.00
	2010	2.23	0.29	0.69	0.20
	2020	5.28	1.30	0.36	0.17

Table 2: Share of children exposed to compound air pollution and heat by group and year

Group	Year	$> 15\ \mu g/m^3$ & > 3 months	$> 15\ \mu g/m^3$ & > 4 months	$> 25\ \mu g/m^3$ & > 3 months	$> 25\ \mu g/m^3$ & > 4 months
HDI *(Q1 = lowest)*					
Q1	2000	3.41	0.17	0.00	0.00
	2010	13.59	3.96	2.32	0.17
	2020	15.67	4.85	2.30	1.20
Q2	2000	4.87	0.34	0.00	0.00
	2010	8.52	5.85	3.50	2.82
	2020	12.38	6.80	3.45	3.21
Q3	2000	0.64	0.00	0.00	0.00
	2010	4.24	0.73	0.89	0.26
	2020	4.68	1.47	0.43	0.19
Q4	2000	0.00	0.00	0.00	0.00
	2010	0.00	0.00	0.00	0.00
	2020	0.00	0.00	0.00	0.00
Urban share *(Q1 = most rural)*					
Q1	2000	9.36	0.00	0.00	0.00
	2010	43.66	16.02	12.67	5.23
	2020	40.46	17.44	9.90	5.78
Q2	2000	9.90	0.00	0.00	0.00
	2010	30.81	15.23	12.89	7.39
	2020	30.46	14.78	9.00	6.67
Q3	2000	4.81	0.00	0.00	0.00
	2010	17.07	7.07	3.05	1.25
	2020	16.26	6.56	2.06	1.31
Q4	2000	1.65	0.16	0.00	0.00
	2010	4.68	1.57	1.05	0.41
	2020	6.90	2.46	1.20	0.89

Note: Share of children (in percent) exposed to both $PM_{2.5}$ above $15\,\mu g/m^3$ (IT3) or $25\,\mu g/m^3$ (IT2) and at least 3 or 4 months per year above 32°C. We obtain it by overlapping cell-level $PM_{2.5}$, UTCI, and child population data. Rows and group definitions are as in Table 1.

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Appendix for
Air Pollution and Extreme Heat Exposures for Young Children in Latin America and the Caribbean

A Materials and Methods

A.1 Conceptual framework

Figure D.1 summarizes how air pollution and extreme heat may affect early childhood development (ECD). It motivates our focus on young children but is not tested in this study; we highlight four features.

First, child development encompasses multiple components, as illustrated in the bottom (gray) box in the figure. We identify five key dimensions: birth outcomes, mental health, physical health, cognitive development, and physical development. These components can interact either contemporaneously or over time (e.g., the dynamic complementarities highlighted by Cunha and Heckman (2008)). Other important aspects of ECD are not included in the framework. For instance, the Nurturing Care Framework includes health, nutrition, learning, relationships, and safety and security (Black et al. 2021). We limit the framework to dimensions for which there is evidence on the effects of air pollution and extreme heat, recognizing that studies addressing all ECD dimensions remain scarce.

Second, child development is a dynamic process with several life-cycle stages within the neonatal, infancy, and toddlerhood phases (ages 0–5). The framework covers impacts from the in-utero period up to age five. The importance of different developmental components may vary across ages within and across these stages, so empirical findings and their implications may also differ by age and stage. Although we do not cover stages beyond age five, shocks experienced by children aged five or younger can significantly affect later stages of development.

Third, extreme heat and air pollution can directly and indirectly affect ECD dimensions. The arrow in Figure D.1 linking the green and gray boxes illustrates the direct effects of exposure on development, such as heat stress and respiratory complications. Indirect effects stem from influences on caregivers and households (blue box) and communities (red box) that may be amplified by vulnerability factors (purple box). For instance, extreme heat can lead to crop failure, resulting in negative income shocks that reduce household investment in children's human capital and nutrition, particularly for children living in greater poverty and with less access to health and other services. These mechanisms may also interact and jointly affect several aspects of child development.

Fourth, extreme heat and air pollution affect child development through the same ecological layers, although the magnitude and specific mechanisms may vary. For example, indirect effects through vector-borne diseases may be significant for extreme heat but are unlikely mechanisms for air pollution.

A.2 Data

In this section, we describe our sources for population, air pollution, and extreme heat data. For a particular subnational, national and supranational region, we compute the child-population-

weighted pollution or heat exposure measures using the distribution of children jointly with the distribution of climatic burdens at the finest level of geography where data are available. Our statistics inform policy makers where a specific population group, in this case children 0 to 5, faces different magnitudes of climatic exposures. Given that the spatial distribution of children in Latin America and the Caribbean might not overlap with the spatial distribution of climatic burdens, it is essential to combine the population distribution and climatic distribution data jointly to identify locations where children are most at risk.

Children population

We generate cell-specific global population estimates based on the WorldPop population estimates dataset for 2000, 2010, and 2020 (Tatem 2017). The WorldPop gridded data provides total population estimates at 30 arc-second grids ($\sim$ 1km at the equator), and it is aggregated based on up to administrative level 6 population data from global economies. Population estimates are provided by age and gender categories, which we restrict to 0-5 years to calculate child population.

Air pollution

Atmospheric pollution by aerosols is important to human health and well-being, especially when there are higher concentrations of $PM_{2.5}$ particles that are smaller than 2.5 micrometers (Jacobson 2002; Bishop, Ketcham, and Kuminoff 2023; Currie et al. 2015; Currie and Walker 2011). We use Donkelaar et al. (2016) estimates available at 0.01° as our measure of air pollution. We chose particulate matter (PM 2.5) as a pollution measure following research indicating that these particles are small enough to enter human bodies and cause health complications (Odo et al. 2023; Fisher et al. 2021; Fu, Viard, and Zhang 2021). We aggregated the data into $1° \times 1°$ grids, calculating the annual average $PM_{2.5}$ level for each grid based on the cells within it. We aggregate these measures to the child population mean exposure for higher geographical groups using cell-level information.

To have benchmarks for air-pollution severity, we use the recommended annual level of PM 2.5 concentration before 2021 — 10 $\mu g/m^3$ — and interim targets (ITs) of the World Health Organization (WHO) as a threshold analysis (*WHO global air quality guidelines* 2021). These thresholds represent staged concentration limits that provide a framework for progressive air quality improvement. Each successive target (IT1 through IT4) corresponds to a significant decrease in air pollution concentrations, allowing for a gradual approach to air quality policies in regions where immediate adherence to the Air Quality Guideline may be technically or economically infeasible. The $PM_{2.5}$ concentrations corresponding to Interim Targets (IT) 4, 3, 2, and 1 are 10, 15, 25, and 35 $\mu g/m^3$, respectively. We use these targets as discrete measures of health-risk exposure to calculate the number of children exposed to pollution levels above and within these thresholds. For example, using population and air pollution data, we compute the population exposed to pollution levels above IT 1, 35 $\mu g/m^3$, and between IT2 and IT 1, 25-35 $\mu g/m^3$.

Temperature

We use the fifth generation of the European Centre for Medium-Range Weather Forecasts (ECMWF) atmospheric reanalyses of global climate: the ERA5-HEAT dataset (Napoli 2020). ERA5-HEAT offers hourly data on numerous climatic variables with a spatial resolution of 0.25 degrees. We utilize the Universal Thermal Climate Index (UTCI), an integrative measure of the human-perceived equivalent temperature, taking into account factors like air temperature, humidity, wind speed, and radiant heat (Bröde et al. 2012; Jendritzky, De Dear, and Havenith 2012; Jendritzky and Höppe 2017).

Following Feng et al. (2024), within a particular span of time at a particular gridded location, we compute Share of Time for the Average Child (STAC) statistics. STAC statistics measuring the share of time exposed to extreme heat for the average child, considering multiple temperature thresholds that capture different magnitudes of heat stress. Our main results focus on 32° as a threshold, but we present findings using 26° and 38° in our appendix. We choose to focus on 32°, the lower bound temperature associated with the category of strong heat stress using UTCI (Bröde et al. 2012). To make the share of time easier to interpret, we transform the share of time into months of exposure per year. We check how many children are exposed to 0 to 1, 1 to 2, 2 to 3, 3 to 4, 4 to 5, and 5+ months of extreme heat. We focus our discussion of results on the number of months above 32°, but we also compute these numbers using 26° and 38° of exposure as temperature thresholds.

GDP per capita

Gross domestic product (GDP) per capita is obtained from the Becker Friedman Institute for Economics Data Studio (Rossi-Hansberg and Zhang 2026). The data provide cell-level annual estimates of GDP per capita for 2012 to 2022, expressed in purchasing power parity (PPP) terms in constant 2021 US dollars. Because estimates are not available before 2012, we use 2020 as the reference year, as it is the only year in the series that coincides with one of our analysis years (2000, 2010, and 2020). We assign cells to income quartiles using their 2020 values and hold these assignments constant across all years.

Human Development Index (HDI)

Human development is measured using the Subnational Human Development Index (SHDI) from the Global Data Lab (Smits and Permanyer 2019). The SHDI measures development on a scale from 0 to 1 incorporating measures in health, education, and standard of living. The dataset provides annual HDI estimates at the national and subnational levels. We use observations corresponding to 2000, 2010, and 2020, matching the years available to our master dataset

We assign HDI values to our $1° \times 1°$ geographic grid using the GDL's subnational boundary shapefiles. Each grid coordinate is spatially matched to a GDL subnational polygon and the subnational HDI of the polygon is then attached to the interior grid point. Where a subnational HDI value is unavailable or a coordinate cannot be matched to a subnational region, we substitute the corresponding country's national HDI for the corresponding year though these

cases were largely relevant for small nations under which the entire country could be considered a subnational unit. Of the 6,540 grid-year observations, 77.4% received subnational HDI estimates, 20.5% received national estimates, and 2.1% remained unmatched.

Accessibility

Accessibility is measured using the 2015 Accessibility to Cities map (Weiss et al. 2018), distributed by the Malaria Atlas Project. The map provides, at around 1 km resolution, the travel time in minutes to the nearest city of at least 50,000 inhabitants. We average travel times over the pixels within each $1° \times 1°$ cell and divide cells into quartiles of this mean travel time across Latin America and the Caribbean. Q1 contains the most remote cells, i.e., longest travel times, and Q4 the most accessible. The map provide a single year 2015 information, which implies that we hold remoteness quartiles constant across 2000, 2010, and 2020.

Land type

Land type is defined using the LandMark database of formally recognized Indigenous and community land rights (LandMark 2026). We classify each $1° \times 1°$ cell as Indigenous if its centroid falls within an Indigenous territory, and as non-indigenous otherwise, yielding two groups. LandMark boundaries are treated as time-invariant, so each cell keeps the same classification in 2000, 2010, and 2020.

Rural-Urban Classification

Rural-urban characteristics are measured using the Global Human Settlement Layer Settlement Model (GHS-SMOD R2023A), developed by the European Commission's Joint Research Centre (Pesaresi et al. 2024). The dataset provides global settlement classifications at approximately 1km resolution using a combination of gridded population estimates and built-up surface information. We use the 2000, 2010, and 2020 settlement layers and aggregate the original, highly detailed classifications into the individual classifications of urban, rural, and water.

To integrate these data into our $1° \times 1°$ geographic grid, we calculate the area of intersection between each grid cell and the underlying 1km raster values that fall within a grid cell. These areas are aggregated by settlement category to obtain three variables representing the proportion of each grid cell classified as urban, rural, or water. All three measures are calculated separately for 2000, 2010, and 2020.

Capturing overlapping pollution and temperature exposures

To capture overlapping exposures to environmental hazards, we merge our cell-level data on air pollution, temperature, and child population. Using the merged dataset, we calculate the number of children exposed to both poor air quality and months with extreme heat. We combine Interim Targets (ITs) of the World Health Organization and months exposed to hot temperatures to define groups. For example, we calculate the number of children exposed to $PM_{2.5}$ above IT2 — 25 $\mu g/m^3$ — and 3 months above 32° in LAC, its subregions, and countries.

Our approach reveals hotspots where children are more vulnerable to both high pollution and high temperatures.

A.3 Statistical framework

This section defines the statistics reported in the main text. The unit of observation is a $1° \times 1°$ cell c in year $t \in \{2000, 2005, 2010, 2015, 2020\}$. Let w_{ct} be the number of children aged 0–5 living in the cell, x_{ct} the annual mean $PM_{2.5}$ concentration ($\mu g/m^3$), and $h^{\kappa}_{ct} \in [0, 1]$ the share of hours in the year with UTCI above $\kappa \in \{26, 32, 38\}$°C. The share h^{κ}_{ct} is the mean of the hourly indicator $\mathbf{1}\{UTCI > \kappa\}$ over all hours of the year, computed on the 0.25° grid and then averaged to the 1° cell, so that $h = 0.25$ corresponds to three months. A group g is a set of cells: a country, a subregion, the whole region, or a land-type, income, HDI, accessibility or urbanicity group. Cells with missing values of x_{ct}, h^{κ}_{ct} or w_{ct}, or with $w_{ct} = 0$, are excluded from the corresponding statistic.

Child-weighted distributions. For a group g and year t, the share of the group's children living in cell $c \in g$ is

$$s_{c,g,t} = \frac{w_{ct}}{\sum_{\hat{c} \in g} w_{\hat{c}t}}, \tag{A.1}$$

so that $\sum_{c \in g} s_{c,g,t} = 1$. For any exposure measure y_{ct} (either x_{ct} or h^{κ}_{ct}), the child-weighted distribution function, which gives the share of children in g exposed to at most z, is

$$F_{gt}(z) = P_{gt}(y \leq z) = \sum_{c \in g} s_{c,g,t}\, \mathbf{1}\{y_{ct} \leq z\}. \tag{A.2}$$

A cell with twice as many children therefore counts twice. The mean of this distribution is

$$\bar{y}_{gt} = \sum_{c \in g} s_{c,g,t}\, y_{ct}. \tag{A.3}$$

With $y = h^{\kappa}$, Equation (A.3) is the Share of Time for the Average Child (STAC) of Feng et al. (2024): the share of the year that the average child in g spends above κ. Weighting by children makes $\bar{y}_{gt}$ describe the conditions of the average child, not of the average cell.

Percentiles exposure and dispersion. We define the τ-th percentile as the smallest exposure level at which at least τ% of children in g are covered:

$$Q^{\tau}_{gt} = \min\{z : F_{gt}(z) \geq \tau/100\}. \tag{A.4}$$

We report the 20th and 80th percentiles, Q^{20}_{gt} and Q^{80}_{gt}, and measure the dispersion of exposure within a group as their difference,

$$D_{gt} = Q^{80}_{gt} - Q^{20}_{gt}, \tag{A.5}$$

in $\mu g/m^3$ for $PM_{2.5}$ and in share of the year for heat. D_{gt} is the amount by which the child at the 80th percentile is more exposed than the child at the 20th.

Children above a threshold. For an interval $[a, b)$ of exposure values, the number of children in group g is

$$N_{gt}[a, b) = \sum_{c \in g} w_{ct}\, \mathbf{1}\{a \leq y_{ct} < b\}, \tag{A.6}$$

and their share is

$$S_{gt}[a, b) = \sum_{c \in g} s_{c,g,t}\, \mathbf{1}\{a \leq y_{ct} < b\} = \frac{N_{gt}[a, b)}{\sum_{c \in g} w_{ct}}. \tag{A.7}$$

For $PM_{2.5}$ the intervals are delimited by the WHO guideline and interim targets (10, 15, 25 and 35 $\mu g/m^3$). For heat the intervals are one-month bins of h^{32}, with edges at $1/12, 2/12, \ldots$ of the year. For instance, the three-month and four-month thresholds are $h = 0.25$ and $h = 0.33$.

Compound exposure. The number of children who face both hazards is

$$J_{gt}(\pi, m) = \sum_{c \in g} w_{ct}\, \mathbf{1}\{x_{ct} > \pi\}\, \mathbf{1}\{h^{32}_{ct} \geq m\}, \tag{A.8}$$

with $\pi \in \{15, 25\}$ $\mu g/m^3$ and $m \in \{0.25, 0.33\}$ (three and four months). Our headline measure is $J_{gt}(15, 0.25)$ and its share, $J_{gt}(15, 0.25)/\sum_{c \in g} w_{ct}$.

Environmental change versus changes in where children live. To separate the two, we recompute each statistic with the 2000 child distribution held fixed:

$$\bar{y}^{CF}_{gt} = \sum_{c \in g} s_{c,g,2000}\, y_{ct}. \tag{A.9}$$

The change in the mean between 2000 and t then splits exactly into

$$\bar{y}_{gt} - \bar{y}_{g,2000} = \underbrace{\left(\bar{y}^{CF}_{gt} - \bar{y}_{g,2000}\right)}_{\text{environmental change}} + \underbrace{\left(\bar{y}_{gt} - \bar{y}^{CF}_{gt}\right)}_{\text{change in where children live}}. \tag{A.10}$$

The first term changes the environment and keeps the 2000 children in place; the second changes the children's locations and keeps the environment at its year-t values. The two sum to the total change with no residual. The distributions in Figure 1 apply the same logic to the full distribution, by replacing the shares $s_{c,g,t}$ with $s_{c,g,2000}$ in Equation (A.2).

Fixed groups. For land type, income, HDI, accessibility and urbanicity, group membership is held fixed over time: quartiles are defined from 2020 values (2015 for accessibility), so that changes within a group reflect changes in exposure and in the children living in its cells, not reclassification of cells across groups.

B Additional results

B.1 LAC region results

Our results show that between 2000 and 2020, the number of children in the region facing lower air quality and increasing heat exposure increased sharply.

We report mean statistics in Table C.1, shares of children above thresholds are reported in Tables C.2 and C.3, and distributional results are visualized in Figure 1, which presents the

distributions of $PM_{2.5}$ and shares of hours (within a year) above 32° for children in years 2000, 2010, and 2020, using cell-level statistics. These distributions are plotted using child population and climatic data aggregated at the cell-level. In the figure, we add vertical lines to indicate Interim Targets and share-of-hours-of-exposure (summarized in months units) thresholds. We complement the visualization with shaded areas to make clear the range of values associated with each severity category.

Air pollution. The mean child air pollution exposure in the region increased from 13.05$\mu g/m^3$ in 2000 to 15.96$\mu g/m^3$ in 2020. In 2020, 40.6 million children (87% of the population) were exposed to air quality that did not meet WHO safety (10μ g/m3), an increase of 7.9 million children compared to 2000. In terms of severity, in 2010, we observed children exposed to levels above 25 and 35 $\mu g/m^3$, the two highest WHO Interim Targets, a new fact relative to 2000, and an increase in the mass of cells exposed to levels above 25 and 15 $\mu g/m^3$. In particular, we observe 200 thousand children in 2020 exposed to levels exceeding the worst severity threshold, a new risk category, as no children were exposed to this level of air pollution in the year 2000.

Heat stress. Focusing on extreme heat, children were more exposed to hotter temperatures in 2020 compared to 2000. The mean number of hours that the average child was exposed to UTCI temperature over 32° in 2000 was around 730 hours (1 month) or 9.5% of the whole year, which increased to 12.5% in 2020. We find that the number of children exposed to at least 2190 hours per year (three months) of temperatures exceeding UTCI 32°C nearly doubled in these two decades, reaching 7.53 million. Considering alternative durations of exposure, we find that there was an increase in children exposed to 3-to-5 months above 32° between 2000 and 2020. Tables C.4 and C.5 show this increase in the number of children exposed to different month thresholds. From 2000 to 2020, an additional 3.5 million children were exposed to 3-to-5 months of temperatures above 32°, representing around 16% of the child population under five years of age.

Joint exposure to air pollution and heat stress. The overlapping analysis of extreme temperature and air pollution in the region reveals an increase in the number of children facing an overlap of these hazards. In 2020, 4.3 million children (9.2%) faced extreme heat (more than 2190 hours, or three months, per year above 32°) and high particulate matter ($PM_{2.5}$ above 15 $\mu g/m^3$, IT3), 3.2 million more than in 2000.

The heat map in Figure D.2 shows the distribution of children across pollution and heat categories. Table C.6 shows that the number of children facing air pollution levels above 15 $\mu g/m^3$ (IT3) and 4 months of exposure to temperatures above 32° rose from 0.06 to 1.66 million between 2000 and 2020. For 25 $\mu g/m^3$ (IT2) and 4 months of exposure to temperatures above 32°, it rose from none to 0.56 million children. Figure D.2 shows this overlap of vulnerabilities in a more granular dimension. From 2000 to 2020, we observed an increase in the child population mass across combinations of temperature thresholds and pollution levels above 25$\mu g/m^3$. Tables C.6 and 2 show that in 2000, there were no children exposed to both air pollution levels above 25 $\mu g/m^3$ (IT2) and more than 3 months of exposure above 32°C, a phenomenon that is present in both 2010 and 2020.

Decomposing environmental and population shifts over time. In this section, we decompose the relative contribution of changes in the distribution of child population and changes in air pollution exposure and heat exposure over time to the overall changes in the number of children exposed to air pollution and heat stress over time. The number of children exposed to heat stress can increase even if there are no changes in the spatial distribution of air pollution and heat, if children over time tend to reside and be born in locations that are more polluted and hotter.

To conduct this analysis, we combine the spatial distribution of pollution and heat exposures in 2010 and 2020 with the spatial distribution of child population in 2000. This allows us to construct counterfactual scenarios in which only the spatial distribution of ambient environmental conditions shifted across time, but population did not.

The results show that the increase in the number of children exposed to air pollution and heat stress was not driven by changes in the spatial distributions of child populations, but driven by increases in air pollution and heat stress across time and over space. We show these distributions on the right plots of Figure 1. We observe that these distributions overlap almost perfectly for both air pollution and extreme heat, which indicates that if the population distribution did not change relative to 2000, the counterfactual child environmental exposure distributions would be similar to the observed distribution.

B.2 LAC subregional results

Air pollution. The increase in mean air pollution exposure is similar between the Caribbean, Central America, and South America, as seen in Table C.1. The average air-pollution exposure increased in all three subregional groups. South America had the highest increase compared to 2000 of $3.6\mu g/m^3$. However, Central America is the most polluted subregion in LAC with an average concentration of $18.1\mu g/m^3$ in 2020.

Taking into account the severity thresholds for pollution in Tables C.2 and C.3, Central America is the region where children are proportionally more exposed to worse air-quality conditions. In 2000, this was the only region with locations above 25 $\mu g/m^3$ (IT2), and in 2020, it was the only one to have children exposed above 35 $\mu g/m^3$ (IT1). In 2020, more than 15% of the child population in Central America lived in places with concentrations above 25 $\mu g/m^3$ (IT2).

We also observe an increase in the inequality of air pollution exposure faced by children in the subregions. For example, in the Caribbean, we find a higher share of children living below 10 $\mu g/m^3$ in 2020, 40% compared to 33% in 2000. However, the proportion of children exposed to over 15 $\mu g/m^3$ increased from 2% to 38% of the child population in 2020. These changes indicate a more unequal subregion in comparison to 2000. Figures D.3 and D.4 also reflect increased inequality, as shown by the percentile distance increase in 2020 compared to 2000.

Heat stress. Similar to the air-pollution results, extreme-heat exposure within subregions and countries increased on average and also exacerbated exposure inequalities relative to 2000. Table C.1 shows higher shares of the year above $32°C$ in all three subregions. The Caribbean had the highest increase, 4%, corresponding to approximately 15 additional days above $32°C$.

Tables C.4 and C.5 indicate that South America is proportionally the worst subregion when considering the number of months of exposure to high temperatures. In 2000, 470 thousand children were exposed to 4-to-5 months of temperatures above $32°C$, and it was the only subregion with child exposure above 4 months. Our data shows that this number increased to 2.32 million in 2020, accounting for 7.59% of the child population.

Similarly to increases in pollution inequality discussed previously, Figures D.3 and D.4 show increasing percentile exposure distances relative to 2000, indicating that the distance between the level of exposure faced by the lowest percentile and the highest percentile is wider. These patterns are consistent across all subregions in Latin America and the Caribbean.

Joint exposure to air pollution and heat stress. Regarding overlapping vulnerabilities, Table C.6 shows that they are present in Central America and South America and increased considerably between 2000 and 2020. In 2000, South America was the only subregion with children exposed to both air-pollution levels above 15 $\mu g/m^3$ (IT3) and more than 4 months of hot temperatures, with 60 thousand children. However, in 2020 this number increased to 1.49 million. In addition, in 2020, 170 thousand Central American children were also exposed to these overlapping vulnerabilities.

B.3 Country-level results

Air pollution. Figures D.5 and D.6 map countries in Latin America and the Caribbean colored according to their weighted mean air pollution exposures. Categories are created based on distribution quintiles in 2000 for each measure to make it easier to see increases in mean exposures over time. Darker colors indicate worse exposures.

Aligned with the rising air pollution level patterns observed at the regional and subregional level, mean air pollution increased in multiple countries in the region, more specifically nations in the south of Central America and South America. Many countries that were in the 1st, 2nd, and 3rd quintiles in 2000, such as Brazil, Peru, Colombia, Guyana, and Suriname, had pollution corresponding to the 5th quintile in 2020.

Heat stress. For temperature, we also observed a transition of countries from lower quintiles to the first quintiles, confirming the increase in hot days discussed before. Specific examples are the south of Central America and Bolivia, Chile, and Paraguay.

The inequality of exposures faced by child populations also increased over time. We also observe an increase in exposure inequalities to air pollution and hot days within countries. Figure D.3 plots three numbers for each country in South America: air pollution exposure for the 20th percentile of the child population (left dot), the 80th percentile (right dot), and population weighted mean exposure (middle dot). Relative to 2000, countries' means shifted to the right for both pollution and heat, indicating higher pollution and temperature exposure levels. Additionally, the distance between p20 and p80 increased for some countries, implying more unequal exposure within these countries.

Moreover, Table C.7 shows that, over time, multiple countries had higher child populations above higher pollution thresholds. For example, Brazil had all of its population concentrated below 15 $\mu g/m^3$ (IT3) before 2010, but in 2020, 400 thousand children were exposed to

pollution levels above 25 $\mu g/m^3$ (IT2). Additionally, concentration in locations following WHO guidelines decreased, which indicates an increase in within-country exposure inequality. Similar patterns are also present in other countries such as Argentina, Bolivia, Honduras, and Peru.

The inequality in extreme heat exposure also increased over time. The majority of countries face increases in the concentration of children above higher thresholds; however, a limited number of countries, such as Peru, Nicaragua, and Venezuela, have children in 2020 concentrated in categories unpopulated in 2000.

Joint exposure to air pollution and heat stress. Overlapping exposures show a concerning increase in double vulnerability in some countries. For instance, Brazil had an increase of 830 thousand children exposed to both air pollution levels above 15 $\mu g/m^3$ (IT3) and at least 4 months of exposure above 32° in 2020 relative to 2000. Another country with a concerning increase is Colombia, where an additional 270 thousand children were exposed to levels above 15 $\mu g/m^3$ (IT3) and 4 months of exposure above 32° in 2020.

C Tables

Table C.1: Mean PM 2.5 and time above temperature thresholds by Admin 1

Region	Year	Mean PM 2.5	time > 26°	time > 32°	time > 38°
LAC	2000	13.05	0.28	0.10	0.01
LAC	2010	16.44	0.31	0.11	0.01
LAC	2020	15.96	0.34	0.13	0.02
Caribbean	2000	10.69	0.46	0.14	0.01
Caribbean	2010	10.10	0.52	0.17	0.01
Caribbean	2020	11.94	0.56	0.18	0.01
Central America	2000	16.91	0.28	0.09	0.01
Central America	2010	17.01	0.29	0.09	0.01
Central America	2020	18.10	0.32	0.11	0.01
South America	2000	11.49	0.28	0.10	0.01
South America	2010	16.48	0.32	0.12	0.01
South America	2020	15.16	0.33	0.13	0.02

Note: This table show the number of children exposed to different pollution thresholds in each subregion in Latin America and the Caribbean. We obtain it by overlapping cell-level $PM_{2.5}$ levels and child population data and categorizing each cell according to the World Health Organization Interim Targets. Who Guideline corresponds to $PM_{2.5}$ levels $< 10\mu g/m^3$, WHO IT4 to $10 - 15\mu g/m^3$, WHO IT3 to $15 - 25\mu g/m^3$, WHO IT2 to $25 - 35\mu g/m^3$, and WHO IT1 to $35 + \mu g/m^3$. We show these numbers for 2000, 2010, and 2020

Table C.2: Number of children exposure to PM 2.5 interim targets in millions by subregion

Region	Year	WHO Guideline	WHO IT4	WHO IT3	WHO IT2	WHO IT1
LAC	2000	12.28	21.09	11.16	0.45	0.00
LAC	2010	6.43	12.97	21.58	3.09	0.18
LAC	2020	6.26	16.44	20.74	3.27	0.21
Caribbean	2000	0.49	0.99	0.00	0.00	0.00
Caribbean	2010	0.78	0.61	0.03	0.00	0.00
Caribbean	2020	0.66	0.36	0.63	0.00	0.00
Central America	2000	1.18	4.10	7.44	0.45	0.00
Central America	2010	1.57	3.73	6.65	1.44	0.00
Central America	2020	1.36	4.42	6.49	2.18	0.21
South America	2000	10.61	16.00	3.72	0.00	0.00
South America	2010	4.07	8.63	14.90	1.65	0.18
South America	2020	4.25	11.67	13.62	1.09	0.00

Note: This table shows the number of children exposed to different pollution thresholds in each subregion in Latin America and the Caribbean. We obtain it by overlapping cell-level $PM_{2.5}$ levels and child population data and categorizing each cell according to the World Health Organization Interim Targets. Who Guideline corresponds to $PM_{2.5}$ levels $< 10\mu g/m^3$, WHO IT4 to $10 - 15\mu g/m^3$, WHO IT3 to $15 - 25\mu g/m^3$, WHO IT2 to $25 - 35\mu g/m^3$, and WHO IT1 to $35 + \mu g/m^3$. We show these numbers for 2000, 2010, and 2020

Table C.3: Share of children exposed to PM 2.5 interim targets by subregion

Region	Year	WHO Guideline	WHO IT-4	WHO IT-3	WHO IT-2	WHO IT-1
LAC	2000	27.30	46.89	24.81	1.00	0.00
LAC	2010	14.53	29.30	48.76	6.99	0.41
LAC	2020	13.35	35.04	44.21	6.97	0.44
Caribbean	2000	32.95	67.05	0.00	0.00	0.00
Caribbean	2010	54.90	42.70	2.40	0.00	0.00
Caribbean	2020	39.88	21.66	38.46	0.00	0.00
Central America	2000	8.99	31.11	56.49	3.41	0.00
Central America	2010	11.76	27.83	49.65	10.76	0.00
Central America	2020	9.27	30.18	44.28	14.87	1.40
South America	2000	34.98	52.76	12.26	0.00	0.00
South America	2010	13.84	29.32	50.61	5.62	0.62
South America	2020	13.87	38.08	44.48	3.57	0.00

Note: This table shows the share of children exposed to different pollution thresholds in each subregion in Latin America and the Caribbean. We obtain it by overlapping cell-level $PM_{2.5}$ levels and child population data and categorizing each cell according to the World Health Organization Interim Targets. Who Guideline corresponds to $PM_{2.5}$ levels $< 10\mu g/m^3$, WHO IT4 to $10 - 15\mu g/m^3$, WHO IT3 to $15 - 25\mu g/m^3$, WHO IT2 to $25 - 35\mu g/m^3$, and WHO IT1 to $35 + \mu g/m^3$. We show these numbers for 2000, 2010, and 2020

Table C.4: Number of children exposed to temperature > 32 ° in millions

Region	Year	0-1 months	1-2 months	2-3 months	3-4 months	4-5 months	5+ months
LAC	2000	24.63	10.40	5.95	3.54	0.47	0.00
LAC	2010	21.92	11.38	5.72	3.22	2.02	0.00
LAC	2020	21.69	10.49	7.21	5.03	2.50	0.00
Caribbean	2000	0.25	0.60	0.63	0.00	0.00	0.00
Caribbean	2010	0.02	0.80	0.61	0.00	0.00	0.00
Caribbean	2020	0.02	0.85	0.68	0.11	0.00	0.00
Central America	2000	8.07	2.36	2.23	0.52	0.00	0.00
Central America	2010	7.55	3.24	2.20	0.40	0.00	0.00
Central America	2020	7.76	2.82	2.41	1.48	0.17	0.00
South America	2000	16.31	7.45	3.09	3.01	0.47	0.00
South America	2010	14.35	7.34	2.91	2.82	2.02	0.00
South America	2020	13.92	6.82	4.12	3.45	2.32	0.00

Note: This table show the number of children exposed to different temperature thresholds in each subregion in Latin America and the Caribbean. We obtain it by overlapping cell-level UTCI and child population data and categorizing each cell according to the the corresponding time in months. We show these numbers for 2000, 2010, and 2020

Table C.5: Share of children exposed to temperature > 32 ° by subregion

Region	Year	0-1 months	1-2 months	2-3 months	3-4 months	4-5 months	5+ months
LAC	2000	54.75	23.12	13.22	7.87	1.04	0.00
LAC	2010	49.53	25.71	12.92	7.28	4.56	0.00
LAC	2020	46.23	22.35	15.37	10.72	5.32	0.00
Caribbean	2000	17.02	40.24	42.74	0.00	0.00	0.00
Caribbean	2010	1.47	55.69	42.84	0.00	0.00	0.00
Caribbean	2020	1.02	51.37	41.21	6.40	0.00	0.00
Central America	2000	61.24	17.90	16.89	3.98	0.00	0.00
Central America	2010	56.38	24.21	16.41	2.99	0.00	0.00
Central America	2020	52.99	19.24	16.46	10.12	1.19	0.00
South America	2000	53.77	24.56	10.19	9.94	1.54	0.00
South America	2010	48.74	24.94	9.88	9.58	6.85	0.00
South America	2020	45.43	22.27	13.46	11.25	7.59	0.00

Note: This table show the share of children exposed to different temperature thresholds in each subregion in Latin America and the Caribbean. We obtain it by overlapping cell-level UTCI and child population data and categorizing each cell according to the the corresponding time in months. We show these numbers for 2000, 2010, and 2020

Table C.6: Number of children exposed to combined air pollution and temperature thresholds by region

Region	Year	> IT3 & > 3 months	> IT3 & > 4 months	> IT2 & > 3 months	> IT2 & > 4 months
LAC	2000	1.08	0.06	0.00	0.00
LAC	2010	3.31	1.26	0.80	0.35
LAC	2020	4.33	1.66	0.78	0.56
Caribbean	2000	0.00	0.00	0.00	0.00
Caribbean	2010	0.00	0.00	0.00	0.00
Caribbean	2020	0.00	0.00	0.00	0.00
Central America	2000	0.41	0.00	0.00	0.00
Central America	2010	0.21	0.00	0.18	0.00
Central America	2020	1.14	0.17	0.30	0.17
South America	2000	0.67	0.06	0.00	0.00
South America	2010	3.10	1.26	0.62	0.35
South America	2020	3.19	1.49	0.48	0.38

Note: This table show the number of children exposed to different combinations of air pollution and temperature thresholds in each subregion in Latin America and the Caribbean. We obtain it by overlapping cell-level air pollution, UTCI and child population data and categorizing each cell according to the the corresponding time in months. We show these numbers for 2000, 2010, and 2020

Table C.7: Number of children exposure to PM 2.5 interim targets in thousands by country

Country	Year	WHO Guideline	WHO IT-4	WHO IT-3	WHO IT-2	WHO IT-1
ARG	2000	463.13	1276.28	859.99	0.00	0.00
ARG	2010	451.51	738.06	1384.37	39.92	0.00
ARG	2020	302.47	721.04	1732.51	6.28	0.00
BLZ	2000	0.00	122.53	0.00	0.00	0.00
BLZ	2010	0.00	47.73	84.67	0.00	0.00
BLZ	2020	0.00	58.95	94.61	0.00	0.00
BOL	2000	0.00	284.76	956.53	0.00	0.00
BOL	2010	0.00	0.00	216.26	873.99	181.71
BOL	2020	0.00	47.21	663.50	660.87	0.00
BRA	2000	8889.33	5168.84	538.03	0.00	0.00
BRA	2010	3529.20	5610.68	3452.99	514.64	0.00
BRA	2020	3691.13	7246.86	1603.29	401.19	0.00
CHL	2000	130.03	332.82	96.66	0.00	0.00
CHL	2010	8.79	225.06	294.91	0.00	0.00
CHL	2020	36.22	270.39	272.47	0.00	0.00
COL	2000	409.74	3017.76	93.20	0.00	0.00
COL	2010	0.00	85.39	3664.38	0.47	0.00
COL	2020	0.00	634.75	3544.21	0.00	0.00
CRI	2000	0.00	139.08	0.00	0.00	0.00
CRI	2010	0.00	0.00	124.89	0.00	0.00
CRI	2020	0.00	0.00	116.04	0.00	0.00
CUB	2000	127.97	157.46	0.00	0.00	0.00
CUB	2010	232.40	0.00	0.00	0.00	0.00
CUB	2020	231.59	0.00	0.00	0.00	0.00
DOM	2000	334.76	791.63	0.00	0.00	0.00
DOM	2010	531.39	609.62	0.00	0.00	0.00
DOM	2020	408.99	356.98	600.62	0.00	0.00
ECU	2000	348.81	811.02	228.11	0.00	0.00
ECU	2010	0.00	0.68	1468.06	0.00	0.00
ECU	2020	0.00	334.57	1237.72	0.00	0.00
FLK	2000	0.00	0.00	0.00	0.00	0.00
FLK	2010	0.00	0.00	0.00	0.00	0.00
FLK	2020	0.00	0.00	0.00	0.00	0.00
GTM	2000	0.00	0.00	1574.74	39.75	0.00
GTM	2010	0.00	39.26	1037.12	608.03	0.00
GTM	2020	0.00	108.17	565.48	1196.09	0.00
GUF	2000	1.65	2.62	0.00	0.00	0.00
GUF	2010	0.00	5.26	0.24	0.00	0.00
GUF	2020	0.00	7.83	0.00	0.00	0.00
GUY	2000	45.00	5.00	0.00	0.00	0.00
GUY	2010	0.00	5.75	52.10	0.00	0.00
GUY	2020	0.00	27.15	53.06	0.00	0.00
HND	2000	24.14	5.80	1058.21	390.17	0.00
HND	2010	0.00	7.11	517.71	832.48	0.00
HND	2020	0.00	9.08	196.80	981.66	205.26
JAM	2000	0.00	42.72	0.00	0.00	0.00
JAM	2010	0.00	0.00	34.29	0.00	0.00
JAM	2020	0.00	0.00	33.39	0.00	0.00
MEX	2000	1101.20	3406.32	4564.65	18.77	0.00

Country	Year	WHO Guideline	WHO IT-4	WHO IT-3	WHO IT-2	WHO IT-1
MEX	2010	1574.02	3462.01	4322.97	0.00	0.00
MEX	2020	1357.54	3773.27	5244.57	0.00	0.00
MTQ	2000	12.45	0.00	0.00	0.00	0.00
MTQ	2010	9.72	0.00	0.00	0.00	0.00
MTQ	2020	8.49	0.00	0.00	0.00	0.00
NIC	2000	59.49	315.60	245.55	0.00	0.00
NIC	2010	0.00	93.50	518.46	0.00	0.00
NIC	2020	0.00	345.38	267.59	0.00	0.00
PAN	2000	0.00	110.11	0.00	0.00	0.00
PAN	2010	0.00	76.40	40.78	0.00	0.00
PAN	2020	0.00	124.53	0.00	0.00	0.00
PER	2000	2.55	1717.74	789.99	0.00	0.00
PER	2010	0.00	101.26	2419.14	58.14	0.00
PER	2020	13.32	297.81	2516.80	0.00	0.00
PRI	2000	12.26	0.00	0.00	0.00	0.00
PRI	2010	10.37	0.00	0.00	0.00	0.00
PRI	2020	8.36	0.00	0.00	0.00	0.00
PRY	2000	75.07	567.40	4.42	0.00	0.00
PRY	2010	0.00	304.30	300.34	0.00	0.00
PRY	2020	0.00	0.00	611.76	23.83	0.00
SUR	2000	2.67	2.02	0.00	0.00	0.00
SUR	2010	0.00	2.27	2.11	0.00	0.00
SUR	2020	0.00	4.58	0.00	0.00	0.00
URY	2000	37.13	1297.06	0.00	0.00	0.00
URY	2010	84.70	1399.34	0.00	0.00	0.00
URY	2020	83.45	1483.28	0.00	0.00	0.00
USA	2000	0.00	0.00	0.00	0.00	0.00
USA	2010	0.00	0.00	0.00	0.00	0.00
USA	2020	0.00	0.00	0.00	0.00	0.00
VEN	2000	202.59	1517.86	152.60	0.00	0.00
VEN	2010	0.00	156.64	1646.74	166.64	0.00
VEN	2020	122.38	590.29	1389.16	0.00	0.00

Note: This table show the number of children exposed to different pollution thresholds in each country in Latin America and the Caribbean. We obtain it by overlapping cell-level PM 2.5 levels and child population data and categorizing each cell according to the World Health Organization Interim Targets. Who Guideline corresponds to PM 2.5 levels $< 10\mu g/m^3$, WHO IT4 to $10 - 15\mu g/m^3$, WHO IT3 to $15 - 25\mu g/m^3$, WHO IT2 to $25 - 35\mu g/m^3$, and WHO IT1 to $35 + \mu g/m^3$. We show these numbers for 2000, 2010, and 2020

Table C.8: PM 2.5 threshold exposure (shares) by group and year

Group	Year	WHO Guideline	WHO IT-4	WHO IT-3	WHO IT-2	WHO IT-1
Land type						
Indigenous	2000	7.29	55.90	36.81	0.00	0.00
	2010	1.43	46.50	43.21	8.86	0.00
	2020	0.99	59.30	36.13	3.59	0.00
Non-indigenous	2000	27.98	46.58	24.40	1.03	0.00
	2010	15.03	28.65	48.97	6.92	0.43
	2020	14.03	33.70	44.65	7.15	0.46
Accessibility *(Q1 = most remote)*						
Q1	2000	21.10	61.40	17.51	0.00	0.00
	2010	0.49	7.39	70.04	22.08	0.00
	2020	0.56	21.59	66.18	11.67	0.00
Q2	2000	20.08	55.25	23.39	1.28	0.00
	2010	6.07	11.32	68.53	14.09	0.00
	2020	6.28	24.39	59.91	9.42	0.00
Q3	2000	25.17	46.77	28.06	0.00	0.00
	2010	14.45	18.45	59.11	7.62	0.38
	2020	13.18	23.54	57.79	5.50	0.00
Q4	2000	28.80	45.73	24.18	1.29	0.00
	2010	15.85	35.22	42.84	5.61	0.48
	2020	14.62	40.37	37.36	7.01	0.64
Income *(Q1 = lowest)*						
Q1	2000	29.43	32.20	35.26	3.10	0.00
	2010	16.63	10.46	53.56	18.02	1.32
	2020	18.53	16.82	43.86	20.78	0.00
Q2	2000	25.72	39.36	34.91	0.00	0.00
	2010	8.11	20.21	67.18	4.42	0.07
	2020	8.90	18.67	68.51	2.42	1.50
Q3	2000	21.61	66.82	11.58	0.00	0.00
	2010	16.41	41.26	41.56	0.76	0.00
	2020	9.64	55.77	34.18	0.40	0.00
Q4	2000	34.99	55.54	8.89	0.58	0.00
	2010	16.65	63.31	18.77	1.27	0.00
	2020	14.84	66.26	18.47	0.42	0.00
HDI *(Q1 = lowest)*						
Q1	2000	14.83	37.60	44.39	3.18	0.00
	2010	4.02	15.15	63.80	15.82	1.20
	2020	3.93	20.95	57.72	16.08	1.33
Q2	2000	37.48	31.90	30.62	0.00	0.00
	2010	33.54	14.84	45.04	6.50	0.09
	2020	34.61	13.56	45.16	6.67	0.00
Q3	2000	42.49	55.55	1.96	0.00	0.00
	2010	13.97	58.82	26.30	0.92	0.00
	2020	12.27	58.63	28.67	0.43	0.00
Q4	2000	14.34	69.63	16.03	0.00	0.00
	2010	9.20	33.13	57.19	0.48	0.00
	2020	4.86	54.75	40.31	0.07	0.00
Urban share *(Q1 = most rural)*						

Table C.8: PM 2.5 threshold exposure (shares) by group and year

Group	Year	WHO Guideline	WHO IT-4	WHO IT-3	WHO IT-2	WHO IT-1
Q1	2000	33.77	47.82	18.41	0.00	0.00
	2010	11.95	18.71	50.04	19.30	0.00
	2020	14.13	28.02	45.77	12.08	0.00
Q2	2000	37.97	39.41	22.62	0.00	0.00
	2010	23.63	18.00	36.22	19.73	2.42
	2020	22.06	21.82	42.39	13.74	0.00
Q3	2000	46.51	41.30	11.18	1.01	0.00
	2010	26.67	27.87	39.51	5.95	0.00
	2020	26.15	31.29	38.94	3.62	0.00
Q4	2000	23.82	48.05	27.08	1.05	0.00
	2010	12.34	30.10	50.67	6.50	0.40
	2020	11.04	36.24	45.06	7.14	0.53

Note:

Table C.9: Temperature threshold exposure (t32, shares) by group and year

Group	Year	0–1 m	1–2 m	2–3 m	3–4 m	4–5 m	5+ m
Land type							
Indigenous	2000	55.01	22.44	10.91	11.57	0.08	0.00
	2010	51.77	21.53	4.62	15.03	7.05	0.00
	2020	60.06	15.90	2.78	15.38	5.88	0.00
Non-indigenous	2000	54.74	23.15	13.30	7.74	1.07	0.00
	2010	49.44	25.87	13.24	6.98	4.46	0.00
	2020	45.47	22.70	16.07	10.47	5.29	0.00
Accessibility *(Q1 = most remote)*							
Q1	2000	21.00	10.05	15.30	53.66	0.00	0.00
	2010	17.95	9.36	7.14	27.91	37.64	0.00
	2020	8.94	14.19	10.63	38.10	28.14	0.00
Q2	2000	53.97	7.19	14.00	22.72	2.12	0.00
	2010	47.04	9.77	10.78	16.46	15.96	0.00
	2020	43.08	11.00	8.07	19.28	18.57	0.00
Q3	2000	48.87	24.68	16.43	9.28	0.75	0.00
	2010	41.62	28.13	16.26	8.29	5.70	0.00
	2020	39.00	20.73	21.40	13.00	5.87	0.00
Q4	2000	57.47	24.56	12.14	4.79	1.05	0.00
	2010	53.18	27.10	12.27	5.41	2.05	0.00
	2020	50.13	24.41	14.41	8.13	2.91	0.00
Income *(Q1 = lowest)*							
Q1	2000	42.16	25.05	17.58	13.36	1.86	0.00
	2010	33.03	27.79	17.59	13.92	7.67	0.00
	2020	29.12	21.27	24.66	15.33	9.62	0.00
Q2	2000	59.78	17.14	13.48	8.03	1.58	0.00
	2010	54.76	21.67	10.76	5.46	7.35	0.00
	2020	51.95	20.21	11.35	9.52	6.96	0.00
Q3	2000	59.92	26.07	9.39	4.55	0.08	0.00
	2010	56.54	28.71	9.69	4.64	0.44	0.00
	2020	49.84	27.56	12.68	8.90	1.02	0.00
Q4	2000	62.96	24.95	9.06	2.92	0.10	0.00
	2010	61.21	23.93	11.91	2.66	0.29	0.00
	2020	62.42	20.44	9.41	6.36	1.37	0.00
HDI *(Q1 = lowest)*							
Q1	2000	45.09	15.91	22.74	13.97	2.28	0.00
	2010	37.37	23.96	16.17	13.56	8.94	0.00
	2020	33.79	18.49	18.18	19.71	9.84	0.00
Q2	2000	44.44	32.33	11.06	10.83	1.35	0.00
	2010	40.69	30.48	14.73	7.99	6.11	0.00
	2020	35.26	27.11	16.51	13.80	7.31	0.00
Q3	2000	56.10	27.80	12.64	3.47	0.00	0.00
	2010	50.73	29.09	15.40	3.95	0.84	0.00
	2020	50.26	26.24	17.91	4.06	1.53	0.00
Q4	2000	81.79	17.17	1.04	0.00	0.00	0.00
	2010	79.72	18.40	1.89	0.00	0.00	0.00
	2020	76.57	17.99	5.44	0.00	0.00	0.00
Urban share *(Q1 = most rural)*							

Table C.9: Temperature threshold exposure (t32, shares) by group and year

Group	Year	0–1 m	1–2 m	2–3 m	3–4 m	4–5 m	5+ m
Q1	2000	31.56	11.36	22.20	34.36	0.52	0.00
	2010	28.43	6.87	15.29	31.39	18.02	0.00
	2020	24.89	7.73	14.20	31.57	21.61	0.00
Q2	2000	35.39	14.65	19.58	29.15	1.24	0.00
	2010	30.47	13.15	16.19	23.04	17.16	0.00
	2020	26.74	11.18	15.33	27.75	18.99	0.00
Q3	2000	35.87	25.99	15.70	22.16	0.28	0.00
	2010	29.39	25.76	16.43	15.18	13.23	0.00
	2020	25.40	22.14	19.38	18.82	14.26	0.00
Q4	2000	58.70	23.14	12.49	4.51	1.15	0.00
	2010	53.61	26.41	12.23	5.17	2.58	0.00
	2020	50.45	23.02	14.78	8.54	3.21	0.00

Note:

Regions

Table C.10: Share of children exposed to PM 2.5 interim targets by subregion

Region	Year	WHO Guideline	WHO IT-4	WHO IT-3	WHO IT-2	WHO IT-1
Caribbean	2000	32.95	67.05	0.00	0.00	0.00
Caribbean	2010	54.90	42.70	2.40	0.00	0.00
Caribbean	2020	39.88	21.66	38.46	0.00	0.00
Central America	2000	8.99	31.11	56.49	3.41	0.00
Central America	2010	11.76	27.83	49.65	10.76	0.00
Central America	2020	9.27	30.18	44.28	14.87	1.40
South America	2000	34.98	52.76	12.26	0.00	0.00
South America	2010	13.84	29.32	50.61	5.62	0.62
South America	2020	13.87	38.08	44.48	3.57	0.00

Table C.11: Share of children exposed to temperature > 32 ° by subregion

Region	Year	0-1 months	1-2 months	2-3 months	3-4 months	4-5 months	5+ months
Caribbean	2000	17.02	40.24	42.74	0.00	0.00	0.00
Caribbean	2010	1.47	55.69	42.84	0.00	0.00	0.00
Caribbean	2020	1.02	51.37	41.21	6.40	0.00	0.00
Central America	2000	61.24	17.90	16.89	3.98	0.00	0.00
Central America	2010	56.38	24.21	16.41	2.99	0.00	0.00
Central America	2020	52.99	19.24	16.46	10.12	1.19	0.00
South America	2000	53.77	24.56	10.19	9.94	1.54	0.00
South America	2010	48.74	24.94	9.88	9.58	6.85	0.00
South America	2020	45.43	22.27	13.46	11.25	7.59	0.00

Country

Table C.12: Mean PM 2.5 and time above temperature thresholds by country

Country	Year	Mean PM 2.5	time > 26°	time > 32°	time > 38°
ARG	2000	13.64	0.14	0.04	0.00
ARG	2010	16.27	0.15	0.05	0.01
ARG	2020	16.16	0.17	0.06	0.01
BLZ	2000	11.82	0.43	0.16	0.00
BLZ	2010	16.00	0.50	0.20	0.02
BLZ	2020	18.55	0.51	0.24	0.03
BOL	2000	17.31	0.14	0.05	0.00
BOL	2010	30.00	0.17	0.06	0.01
BOL	2020	24.53	0.20	0.08	0.01
BRA	2000	9.42	0.34	0.13	0.01
BRA	2010	13.45	0.39	0.16	0.02
BRA	2020	12.17	0.41	0.17	0.02
CHL	2000	12.19	0.08	0.00	0.00
CHL	2010	15.52	0.08	0.01	0.00
CHL	2020	13.90	0.11	0.02	0.00
COL	2000	12.77	0.28	0.08	0.01
COL	2010	19.48	0.34	0.10	0.01
COL	2020	17.92	0.38	0.13	0.02
CRI	2000	13.19	0.62	0.14	0.00
CRI	2010	17.44	0.63	0.17	0.00
CRI	2020	16.51	0.69	0.19	0.00
CUB	2000	10.25	0.45	0.18	0.02
CUB	2010	8.26	0.48	0.18	0.02
CUB	2020	8.77	0.55	0.22	0.03
DOM	2000	10.78	0.46	0.14	0.00
DOM	2010	10.37	0.53	0.17	0.00
DOM	2020	12.44	0.56	0.18	0.01
ECU	2000	12.91	0.22	0.05	0.00
ECU	2010	19.62	0.27	0.07	0.00
ECU	2020	16.85	0.28	0.09	0.00
FLK	2000	2.71	0.00	0.00	0.00
FLK	2010	3.66	0.00	0.00	0.00
FLK	2020	3.89	0.00	0.00	0.00
GTM	2000	20.62	0.30	0.09	0.00
GTM	2010	23.00	0.33	0.10	0.01
GTM	2020	24.05	0.36	0.12	0.01
GUF	2000	10.38	0.45	0.21	0.00
GUF	2010	13.63	0.55	0.26	0.01
GUF	2020	12.17	0.52	0.24	0.01
GUY	2000	9.51	0.50	0.23	0.01
GUY	2010	19.13	0.60	0.28	0.02
GUY	2020	14.67	0.59	0.28	0.02
HND	2000	21.39	0.40	0.14	0.01
HND	2010	26.31	0.42	0.16	0.01
HND	2020	30.29	0.45	0.18	0.01
JAM	2000	13.47	0.45	0.08	0.00
JAM	2010	15.74	0.55	0.11	0.00

Country	Year	Mean PM 2.5	time > 26°	time > 32°	time > 38°
JAM	2020	16.10	0.57	0.12	0.00
MEX	2000	15.90	0.24	0.07	0.01
MEX	2010	14.58	0.23	0.07	0.01
MEX	2020	15.59	0.28	0.09	0.01
MTQ	2000	6.04	0.34	0.01	0.00
MTQ	2010	6.90	0.51	0.04	0.00
MTQ	2020	6.44	0.46	0.04	0.00
NIC	2000	13.77	0.43	0.14	0.01
NIC	2010	17.70	0.52	0.18	0.02
NIC	2020	15.93	0.52	0.19	0.02
PAN	2000	13.52	0.51	0.12	0.00
PAN	2010	14.18	0.59	0.16	0.00
PAN	2020	13.67	0.63	0.18	0.00
PER	2000	14.25	0.12	0.03	0.00
PER	2010	20.32	0.15	0.04	0.00
PER	2020	18.30	0.16	0.04	0.00
PRI	2000	6.75	0.33	0.03	0.00
PRI	2010	6.21	0.47	0.06	0.00
PRI	2020	6.78	0.44	0.05	0.00
PRY	2000	13.03	0.32	0.16	0.03
PRY	2010	13.89	0.32	0.14	0.03
PRY	2020	21.77	0.37	0.19	0.04
SUR	2000	8.25	0.47	0.22	0.00
SUR	2010	15.28	0.57	0.27	0.01
SUR	2020	12.53	0.55	0.26	0.01
URY	2000	12.24	0.12	0.03	0.00
URY	2010	11.17	0.13	0.03	0.00
URY	2020	11.54	0.14	0.04	0.00
USA	2000	0.00	0.00	0.00	0.00
USA	2010	0.00	0.00	0.00	0.00
USA	2020	0.00	0.00	0.00	0.00
VEN	2000	12.47	0.45	0.18	0.02
VEN	2010	20.09	0.56	0.23	0.03
VEN	2020	16.18	0.55	0.24	0.05

Note: This table show population weighted mean for countries in Latin America and the Caribbean. We show means for air pollution and temperature exposure for 2000, 2010 and 2020.

Table C.13: Number of children exposed to temperature > 32 ° in millions by country

Country	Year	0-1 months	1-2 months	2-3 months	3-4 months	4-5 months	5+ months
ARG	2000	2219.10	380.29	0.00	0.00	0.00	0.00
ARG	2010	2032.46	576.62	4.78	0.00	0.00	0.00
ARG	2020	1983.82	719.04	59.43	0.00	0.00	0.00
BLZ	2000	0.00	40.16	82.37	0.00	0.00	0.00
BLZ	2010	0.00	0.00	132.40	0.00	0.00	0.00
BLZ	2020	0.00	0.00	153.56	0.00	0.00	0.00
BOL	2000	893.64	260.40	53.74	33.52	0.00	0.00
BOL	2010	823.62	287.67	98.92	61.74	0.00	0.00
BOL	2020	855.72	62.56	332.74	108.99	11.58	0.00
BRA	2000	5156.82	5575.35	1462.22	2107.41	294.39	0.00
BRA	2010	3830.00	4588.24	1769.60	1299.55	1620.13	0.00
BRA	2020	3346.99	4196.38	2228.27	1546.39	1624.42	0.00
CHL	2000	559.51	0.00	0.00	0.00	0.00	0.00
CHL	2010	528.76	0.00	0.00	0.00	0.00	0.00
CHL	2020	579.08	0.00	0.00	0.00	0.00	0.00
COL	2000	2383.99	297.11	494.56	174.99	170.06	0.00
COL	2010	2284.51	417.95	390.03	462.70	195.05	0.00
COL	2020	2341.95	556.93	138.37	837.08	304.62	0.00
CRI	2000	0.00	139.08	0.00	0.00	0.00	0.00
CRI	2010	0.00	124.89	0.00	0.00	0.00	0.00
CRI	2020	0.00	0.00	116.04	0.00	0.00	0.00
CUB	2000	1.14	126.17	158.12	0.00	0.00	0.00
CUB	2010	0.86	103.09	128.45	0.00	0.00	0.00
CUB	2020	0.00	47.41	78.66	105.52	0.00	0.00
DOM	2000	183.24	469.08	474.07	0.00	0.00	0.00
DOM	2010	0.00	657.77	483.24	0.00	0.00	0.00
DOM	2020	0.00	765.97	600.62	0.00	0.00	0.00
ECU	2000	1084.26	303.22	0.47	0.00	0.00	0.00
ECU	2010	925.37	444.39	98.30	0.68	0.00	0.00
ECU	2020	594.27	723.49	253.63	0.91	0.00	0.00
FLK	2000	0.00	0.00	0.00	0.00	0.00	0.00
FLK	2010	0.00	0.00	0.00	0.00	0.00	0.00
FLK	2020	0.00	0.00	0.00	0.00	0.00	0.00
GTM	2000	911.75	569.29	23.91	109.54	0.00	0.00
GTM	2010	878.01	652.38	114.76	39.26	0.00	0.00
GTM	2020	944.27	56.97	692.70	175.79	0.00	0.00
GUF	2000	0.00	0.00	4.27	0.00	0.00	0.00
GUF	2010	0.00	0.00	1.82	3.68	0.00	0.00
GUF	2020	0.00	0.00	6.38	1.45	0.00	0.00
GUY	2000	2.48	2.60	14.26	30.66	0.00	0.00
GUY	2010	1.25	4.90	1.84	49.85	0.00	0.00
GUY	2020	3.26	9.08	1.86	21.14	44.87	0.00
HND	2000	720.84	24.14	521.82	211.51	0.00	0.00
HND	2010	215.34	514.29	451.34	176.33	0.00	0.00
HND	2020	0.00	731.06	487.14	0.00	174.60	0.00
JAM	2000	42.72	0.00	0.00	0.00	0.00	0.00
JAM	2010	0.00	34.29	0.00	0.00	0.00	0.00
JAM	2020	0.00	33.39	0.00	0.00	0.00	0.00
MEX	2000	6175.89	1378.03	1333.46	203.55	0.00	0.00

Country	Year	0-1 months	1-2 months	2-3 months	3-4 months	4-5 months	5+ months
MEX	2010	6454.53	1520.26	1229.65	154.57	0.00	0.00
MEX	2020	6816.17	1640.09	829.08	1090.04	0.00	0.00
MTQ	2000	12.45	0.00	0.00	0.00	0.00	0.00
MTQ	2010	9.72	0.00	0.00	0.00	0.00	0.00
MTQ	2020	8.49	0.00	0.00	0.00	0.00	0.00
NIC	2000	259.98	101.84	258.81	0.00	0.00	0.00
NIC	2010	0.00	317.47	263.79	30.70	0.00	0.00
NIC	2020	0.00	347.51	49.93	215.54	0.00	0.00
PAN	2000	0.00	105.47	4.64	0.00	0.00	0.00
PAN	2010	0.00	111.94	5.24	0.00	0.00	0.00
PAN	2020	0.00	42.04	82.48	0.00	0.00	0.00
PER	2000	2273.85	46.27	70.19	119.97	0.00	0.00
PER	2010	2268.76	83.01	81.30	60.41	85.06	0.00
PER	2020	2471.09	93.15	98.90	136.05	28.74	0.00
PRI	2000	12.26	0.00	0.00	0.00	0.00	0.00
PRI	2010	10.37	0.00	0.00	0.00	0.00	0.00
PRI	2020	8.36	0.00	0.00	0.00	0.00	0.00
PRY	2000	0.00	247.31	399.58	0.00	0.00	0.00
PRY	2010	0.00	559.23	45.41	0.00	0.00	0.00
PRY	2020	0.00	143.11	487.45	5.03	0.00	0.00
SUR	2000	0.00	0.00	4.69	0.00	0.00	0.00
SUR	2010	0.00	0.00	0.54	3.84	0.00	0.00
SUR	2020	0.00	0.00	0.43	4.15	0.00	0.00
URY	2000	1334.20	0.00	0.00	0.00	0.00	0.00
URY	2010	1484.03	0.00	0.00	0.00	0.00	0.00
URY	2020	1566.73	0.00	0.00	0.00	0.00	0.00
USA	2000	0.00	0.00	0.00	0.00	0.00	0.00
USA	2010	0.00	0.00	0.00	0.00	0.00	0.00
USA	2020	0.00	0.00	0.00	0.00	0.00	0.00
VEN	2000	400.44	335.17	587.10	547.11	3.23	0.00
VEN	2010	173.67	381.02	418.15	879.18	118.00	0.00
VEN	2020	174.07	318.58	515.24	784.46	309.49	0.00

Note: This table show the number of children exposed to different temperature thresholds in each country in Latin America and the Caribbean. We obtain it by overlapping cell-level UTCI and child population data and categorizing each cell according to the the corresponding time in months. We show these numbers for 2000, 2010, and 2020

Table C.14: Number of children exposed to combined air pollution and temperature thresholds by country

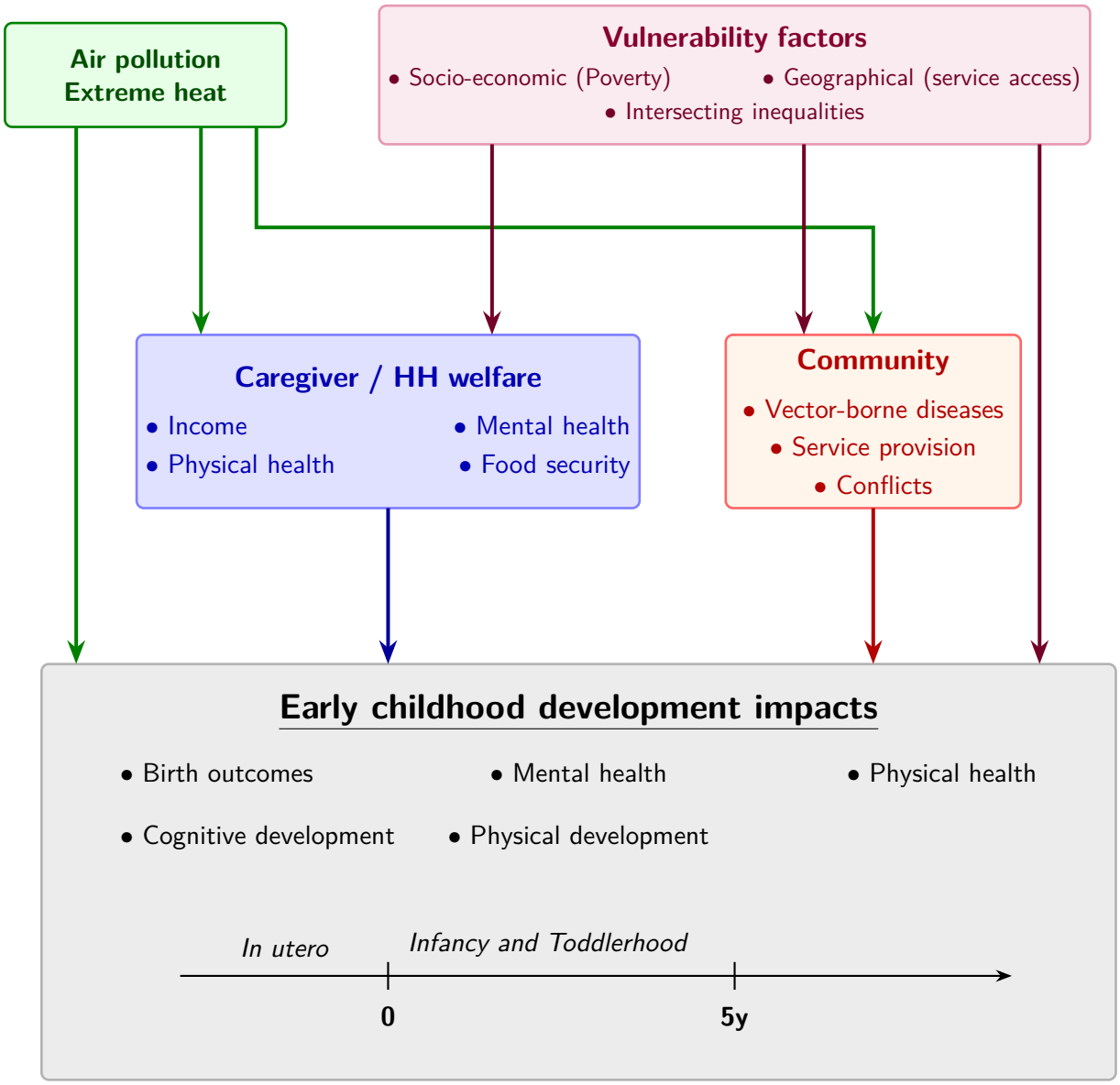
Country	Year	> IT3 & > 3 months	> IT3 & > 4 month	> IT2 & > 3 month	> IT2 & > 4 month
ARG	2000	0.00	0.00	0.00	0.00
ARG	2010	0.00	0.00	0.00	0.00
ARG	2020	0.00	0.00	0.00	0.00
BLZ	2000	0.00	0.00	0.00	0.00
BLZ	2010	0.00	0.00	0.00	0.00
BLZ	2020	0.00	0.00	0.00	0.00
BOL	2000	25.42	0.00	0.00	0.00
BOL	2010	61.74	0.00	52.56	0.00
BOL	2020	120.57	11.58	75.47	11.58
BRA	2000	531.36	36.66	0.00	0.00
BRA	2010	1296.41	866.09	511.66	335.84
BRA	2020	1256.27	884.46	401.19	370.75
CHL	2000	0.00	0.00	0.00	0.00
CHL	2010	0.00	0.00	0.00	0.00
CHL	2020	0.00	0.00	0.00	0.00
COL	2000	23.96	23.96	0.00	0.00
COL	2010	572.36	195.05	0.47	0.47
COL	2020	916.43	304.62	0.00	0.00
CRI	2000	0.00	0.00	0.00	0.00
CRI	2010	0.00	0.00	0.00	0.00
CRI	2020	0.00	0.00	0.00	0.00
CUB	2000	0.00	0.00	0.00	0.00
CUB	2010	0.00	0.00	0.00	0.00
CUB	2020	0.00	0.00	0.00	0.00
DOM	2000	0.00	0.00	0.00	0.00
DOM	2010	0.00	0.00	0.00	0.00
DOM	2020	0.00	0.00	0.00	0.00
ECU	2000	0.00	0.00	0.00	0.00
ECU	2010	0.00	0.00	0.00	0.00
ECU	2020	0.91	0.00	0.00	0.00
FLK	2000	0.00	0.00	0.00	0.00
FLK	2010	0.00	0.00	0.00	0.00
FLK	2020	0.00	0.00	0.00	0.00
GTM	2000	109.54	0.00	0.00	0.00
GTM	2010	0.00	0.00	0.00	0.00
GTM	2020	124.59	0.00	124.59	0.00
GUF	2000	0.00	0.00	0.00	0.00
GUF	2010	0.00	0.00	0.00	0.00
GUF	2020	0.00	0.00	0.00	0.00
GUY	2000	0.00	0.00	0.00	0.00
GUY	2010	44.11	0.00	0.00	0.00
GUY	2020	47.11	43.38	0.00	0.00
HND	2000	211.51	0.00	0.00	0.00
HND	2010	176.33	0.00	176.33	0.00
HND	2020	174.60	174.60	174.60	174.60
JAM	2000	0.00	0.00	0.00	0.00

Country	Year	> IT3 & > 3 months	> IT3 & > 4 month	> IT2 & > 3 month	> IT2 & > 4 month
JAM	2010	0.00	0.00	0.00	0.00
JAM	2020	0.00	0.00	0.00	0.00
MEX	2000	88.50	0.00	0.00	0.00
MEX	2010	0.00	0.00	0.00	0.00
MEX	2020	624.95	0.00	0.00	0.00
MTQ	2000	0.00	0.00	0.00	0.00
MTQ	2010	0.00	0.00	0.00	0.00
MTQ	2020	0.00	0.00	0.00	0.00
NIC	2000	0.00	0.00	0.00	0.00
NIC	2010	30.70	0.00	0.00	0.00
NIC	2020	215.54	0.00	0.00	0.00
PAN	2000	0.00	0.00	0.00	0.00
PAN	2010	0.00	0.00	0.00	0.00
PAN	2020	0.00	0.00	0.00	0.00
PER	2000	5.20	0.00	0.00	0.00
PER	2010	143.00	85.06	0.83	0.00
PER	2020	156.77	28.74	0.00	0.00
PRI	2000	0.00	0.00	0.00	0.00
PRI	2010	0.00	0.00	0.00	0.00
PRI	2020	0.00	0.00	0.00	0.00
PRY	2000	0.00	0.00	0.00	0.00
PRY	2010	0.00	0.00	0.00	0.00
PRY	2020	5.03	0.00	5.03	0.00
SUR	2000	0.00	0.00	0.00	0.00
SUR	2010	1.57	0.00	0.00	0.00
SUR	2020	0.00	0.00	0.00	0.00
URY	2000	0.00	0.00	0.00	0.00
URY	2010	0.00	0.00	0.00	0.00
URY	2020	0.00	0.00	0.00	0.00
USA	2000	0.00	0.00	0.00	0.00
USA	2010	0.00	0.00	0.00	0.00
USA	2020	0.00	0.00	0.00	0.00
VEN	2000	84.14	0.00	0.00	0.00
VEN	2010	985.31	118.00	57.68	13.77
VEN	2020	683.27	216.70	0.00	0.00

Note: This table show the number of children exposed to different combinations of air pollution and temperature thresholds in each country in Latin America and the Caribbean. We obtain it by overlapping cell-level air pollution, UTCI and child population data and categorizing each cell according to the the corresponding time in months. We show these numbers for 2000, 2010, and 2020

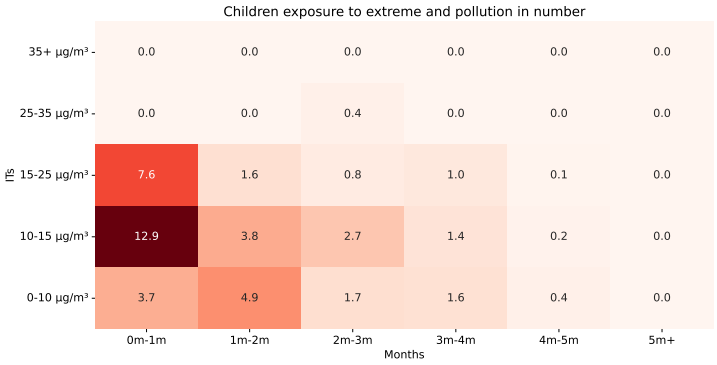
D Figures

Figure D.1: Environment and children framework sketch

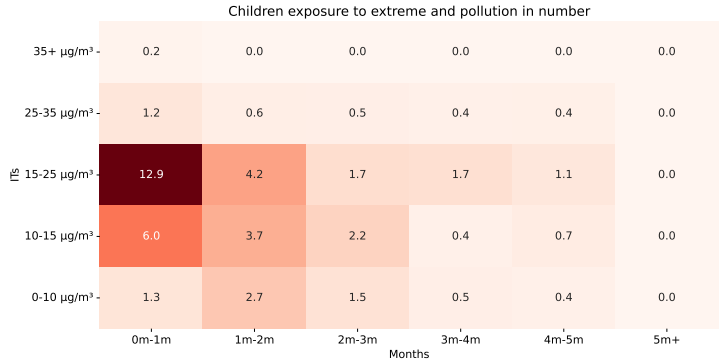


Note: This framework sketch focuses on extreme heat, air pollution exposures, and early childhood development (ECD) from in utero to age 5. The green box represents environmental factors, specifically air pollution and extreme heat exposure, that influence ECD directly and indirectly. The purple box captures vulnerability factors that may affect children's development and the impacts of environmental factors. The blue box contains mechanisms related to caregivers and household welfare that impact ECD. The red box highlights community-level mechanisms affecting ECD outcomes. The gray box represents various dimensions of children's development influenced by the factors and mechanisms outlined in the boxes above.

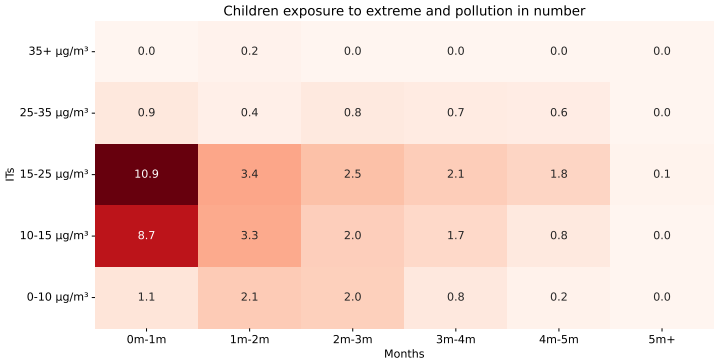
Figure D.2: Children exposed to combined air pollution and heat thresholds in LAC



2000



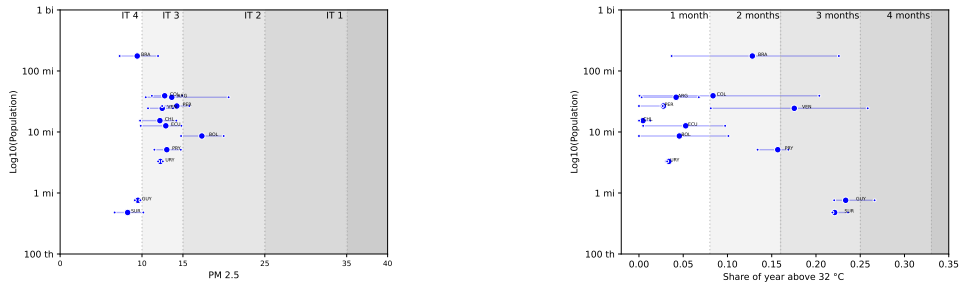
2010



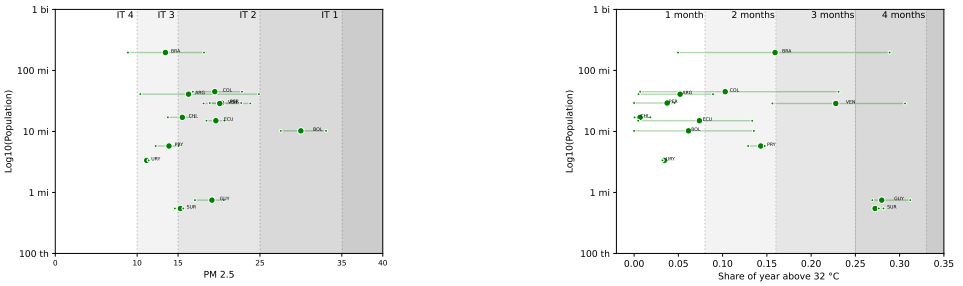
2020

Notes: These heat maps shows the number of children in millions exposed to both pollution and temperature threshold in 2000, 2010, and 2020.

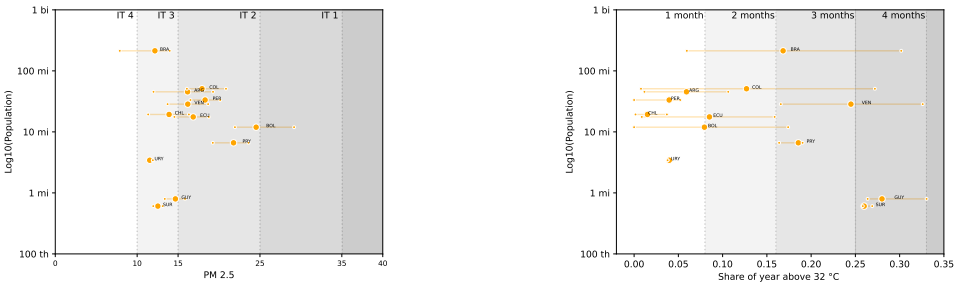
Figure D.3: Country-specific distributional ranges. P20 (left-dot), mean (center-dot), P80 (right-dot)



2000



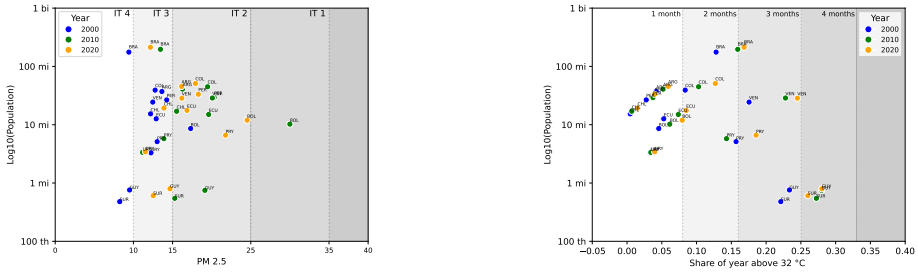
2010



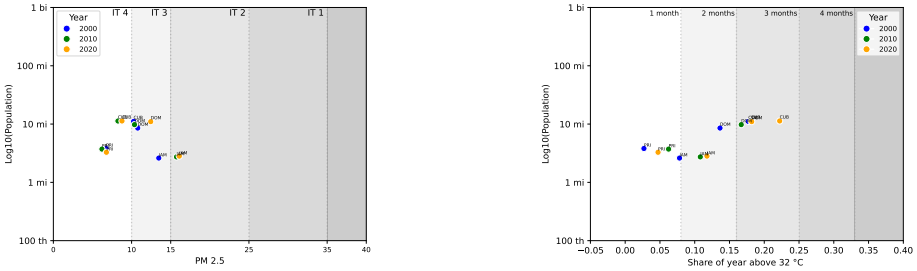
2020

Notes: Lines mark the 20th percentile, mean, and the 80th percentile of a country's $PM_{2.5}$. distribution and share of time above 32 ° , computed based on the distribution of $PM_{2.5}$ levels, time above 32 ° and population across cells corresponding to each country. The x-axis corresponds to levels of $PM_{2.5}$ levels. The vertical dashed lines corresponding to $PM_{2.5}$ thresholds in $\mu g/m^3$ units according to WHO interim targets (ITs) and number of months of extreme heat. Background color corresponds to the IT and month ranges, with darker colors indicating lower air quality thresholds and more time of exposure.

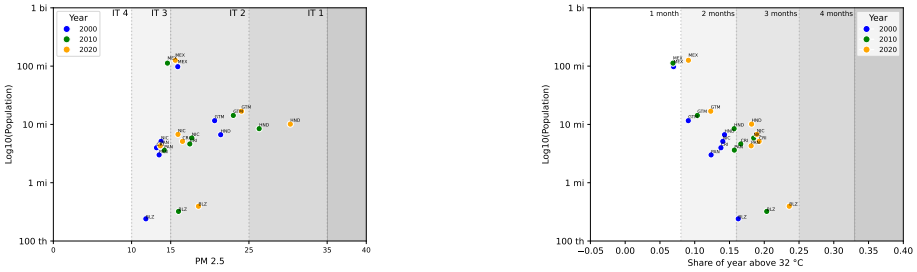
Figure D.4: $PM_{2.5}$ and Share of year above 32 °C Country-specific means by subregion



South America



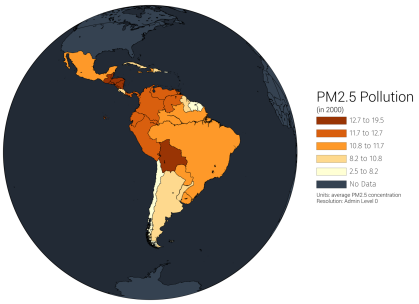
Caribbean



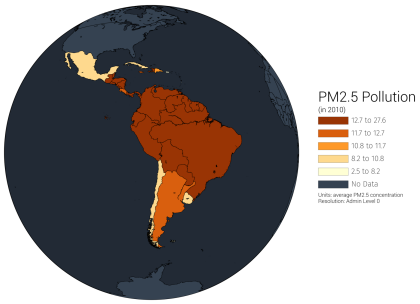
Central America

Notes: Mean country's $PM_{2.5}$ levels and share of year above 32 ° are plotted by subregion. These measures are computed based on the distribution of $PM_{2.5}$ levels, share of time above 32 ° and population across cells corresponding to each country. The x-axis corresponds to levels of $PM_{2.5}$ levels. The vertical dashed lines corresponding to $PM_{2.5}$ thresholds in $\mu g/m^3$ units according to WHO interim targets (ITs) and number of months of extreme heat. Background color corresponds to the IT and month ranges, with darker colors indicating lower air quality thresholds and more time of exposure.

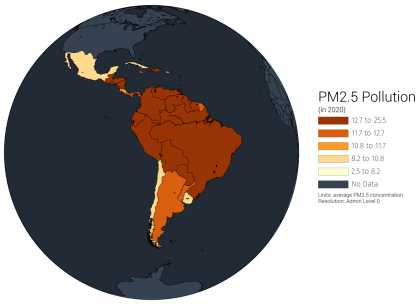
Figure D.5: Country-level $PM_{2.5}$ levels categories



(a) 2000



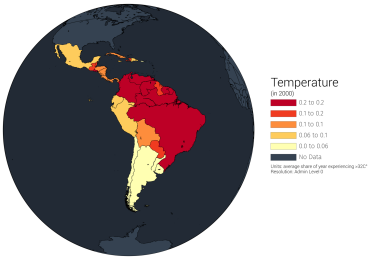
(b) 2010



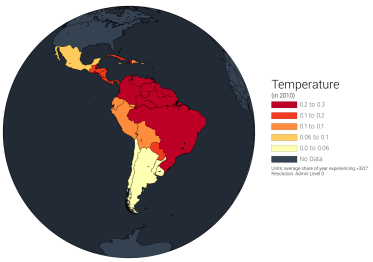
(c) 2020

Notes: These maps display country-level air pollution level aggregated to the country level obtained by overlapping Worldpop data on population and Van Donkelaar et al. (2021) dataset on $PM_{2.5}$ for years 2000, 2010, and 2020. Lighter colors indicate countries with smaller air pollution levels and darker colors indicates more air pollution. Black is used to indicate missing on country-level air pollution information.

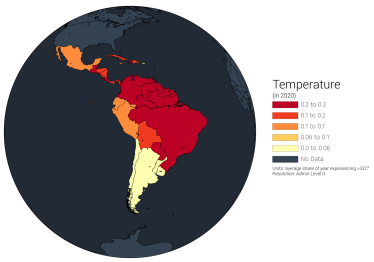
Figure D.6: Country-level share of time > 32°



(a) 2000



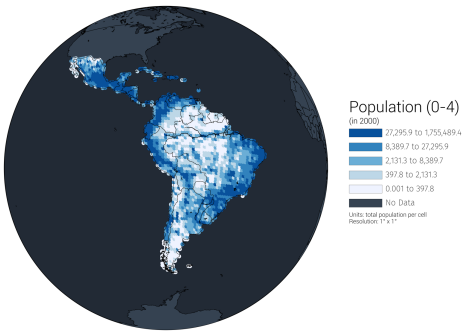
(b) 2010



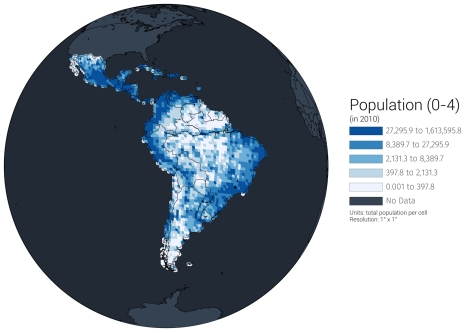
(c) 2020

Notes: These maps display country-level share of year above 32 ° obtained by overlapping Worldpop data on population and UTCI dataset for years 2000, 2010, and 2020. Lighter colors indicate countries with less time above 32 ° and darker colors indicates more time above this temperature. Black is used to indicate missing on country-level temperature information.

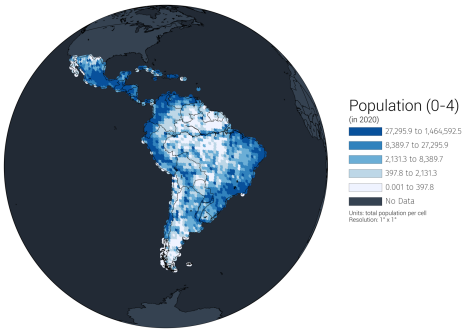
Figure D.7: Cell-level population estimates over $1° \times 1°$ longitude–latitude grids



(a) 2000



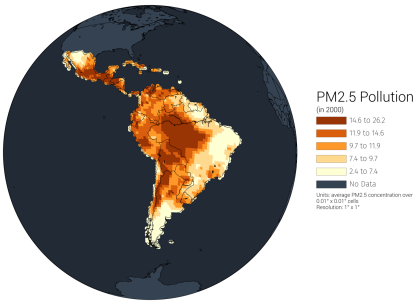
(b) 2010



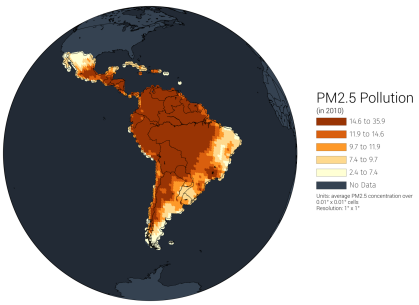
(c) 2020

Notes: These maps display cell-level population estimates in $1^o \times 1^o$ resolution using Worldpop data for years 2000, 2005, 2010, 2015, and 2020. Lighter colors indicate less populated cells and darker colors indicates more populated cells. Black is used to indicate missing on cell-level population estimates, cell country label, and boundaries belonging to regions or countries outside LAC.

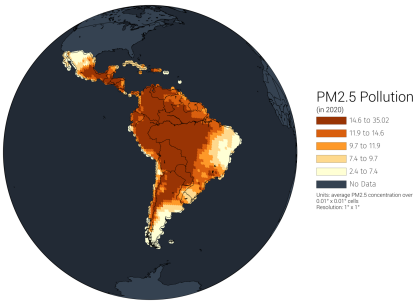
Figure D.8: Cell-level $PM_{2.5}$ level over $1° \times 1°$ longitude–latitude grids



(a) 2000



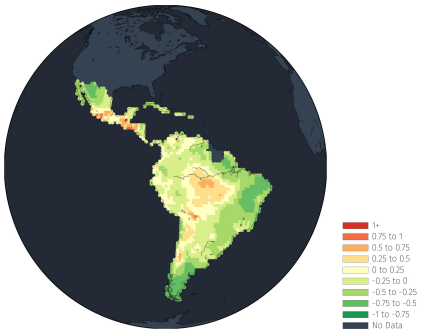
(b) 2010



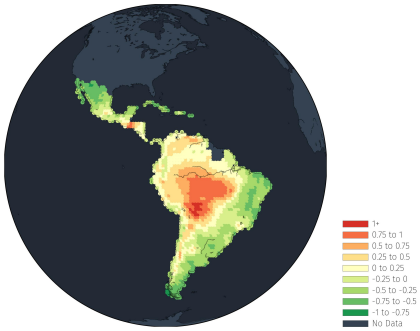
(c) 2020

Notes: These maps display cell-level $PM_{2.5}$ levels in $1^o \times 1^o$ resolution using Van Donkelaar et al. (2021) dataset for years 2000, 2005, 2010, 2015, and 2020. Lighter colors indicate lower levels of air pollution levels and darker colors indicates more higher levels of pollution. Black is used to indicate missing on cell-level air pollution level, cell country label, and boundaries belonging to regions or countries outside LAC.

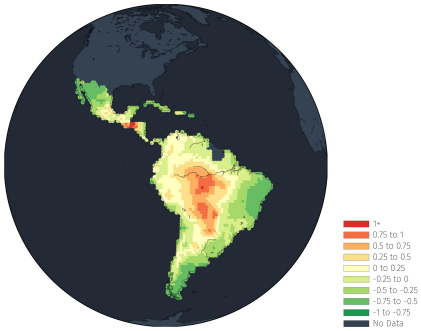
Figure D.9: Cell-level excess burden over 1° × 1° longitude–latitude grids



(a) 2000



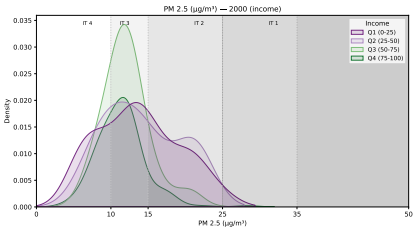
(b) 2010

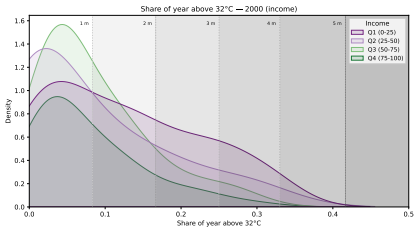


(c) 2020

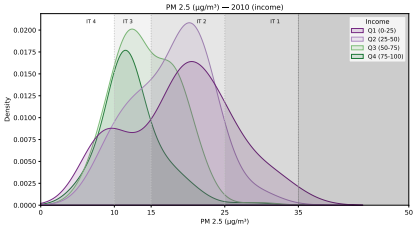
Notes: These maps display cell-level air pollution burden in $1^o \times 1^o$ resolution obtained by overlapping Worldpop data on population and Van Donkelaar et al. (2021) dataset on $PM_{2.5}$ for years 2000, 2005, 2010, 2015, and 2020. Lighter colors indicate cells with smaller air pollution burden and darker colors indicates cells with more air pollution burden. Black is used to indicate missing on cell-level air pollution burden estimates, cell country label, and boundaries belonging to regions or countries outside LAC.

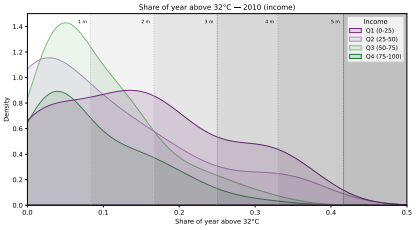
Figure D.10: The distribution of air pollution exposure and heat exposure for children by income group



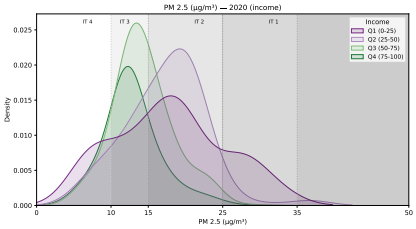


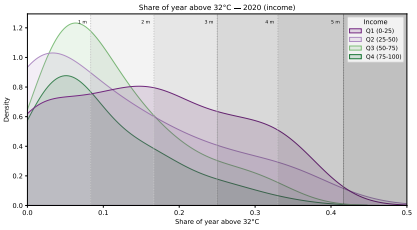
2000





2010

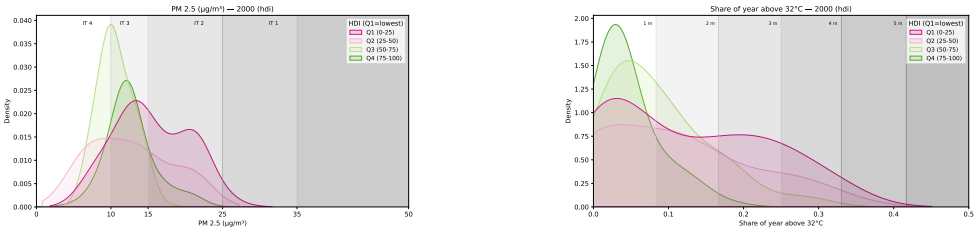




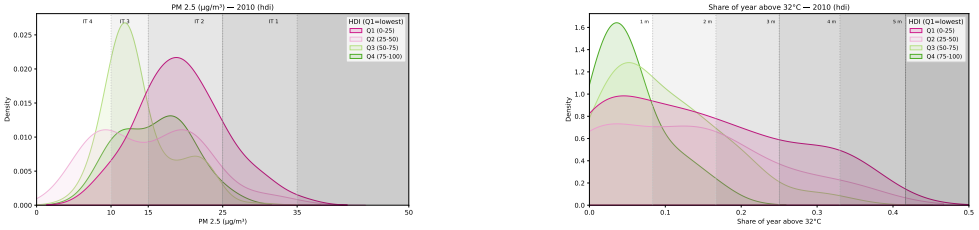
2020

Notes: Kernel density distributions of cell-level $PM_{2.5}$ levels and share of time above 32°C for 2000, 2010, and 2020, weighted by child population. Groups are defined by income quartile based on 2020 GDP per capita at PPP (2021 constant prices), held constant across years. Background colors indicate severity ranges and vertical lines mark severity thresholds.

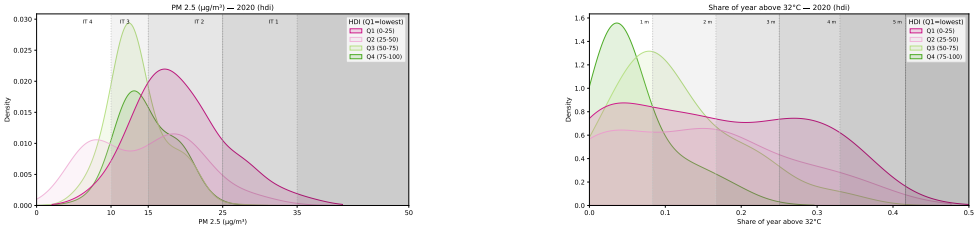
Figure D.11: The distribution of air pollution exposure and heat exposure for children by HDI group



2000



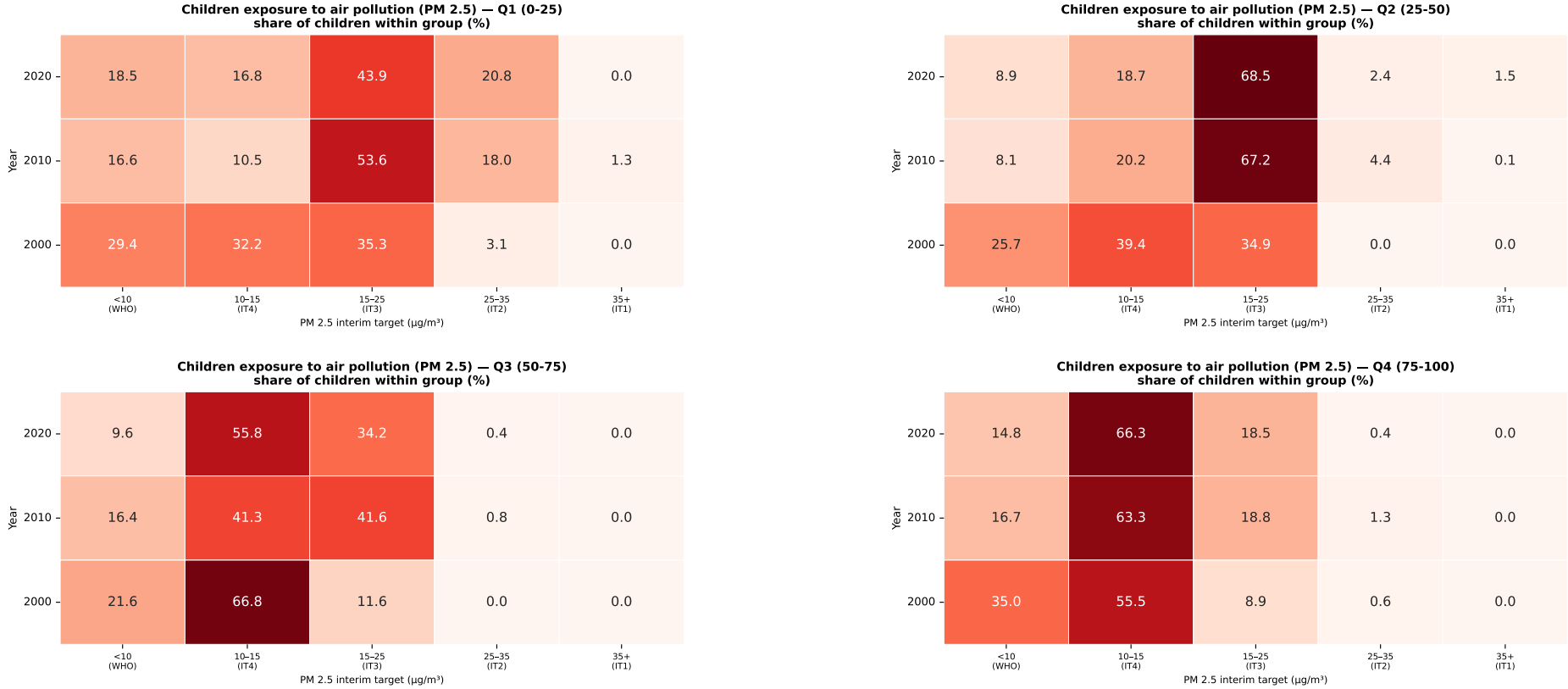
2010



2020

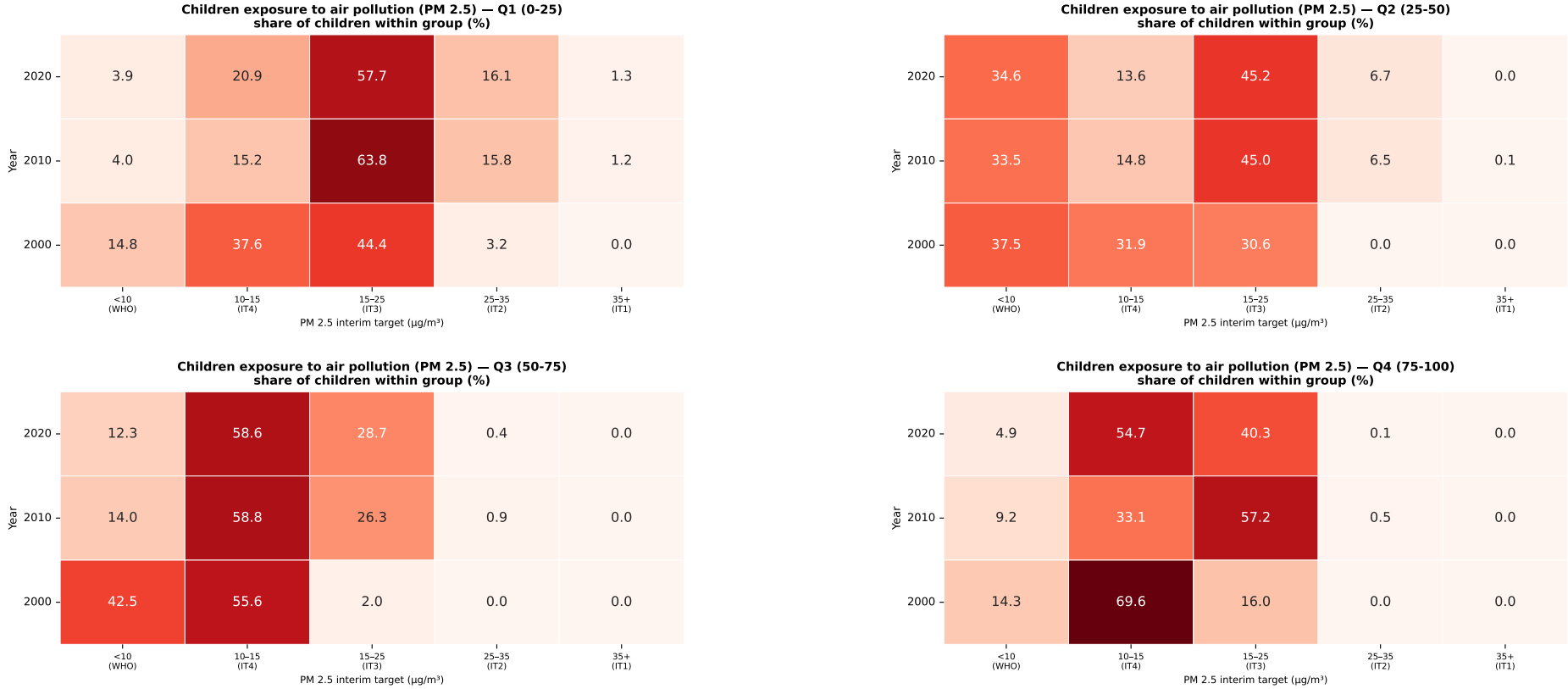
Notes: Kernel density distributions of cell-level $PM_{2.5}$ levels and share of time above 32°C for 2000, 2010, and 2020, weighted by child population. Groups are defined by HDI quartile based on 2020 subnational Human Development Index values, held constant across years. Q1 corresponds to the lowest HDI and Q4 to the highest. Background colors indicate severity ranges and vertical lines mark severity thresholds.

Figure D.12: Children's air pollution exposure by income group



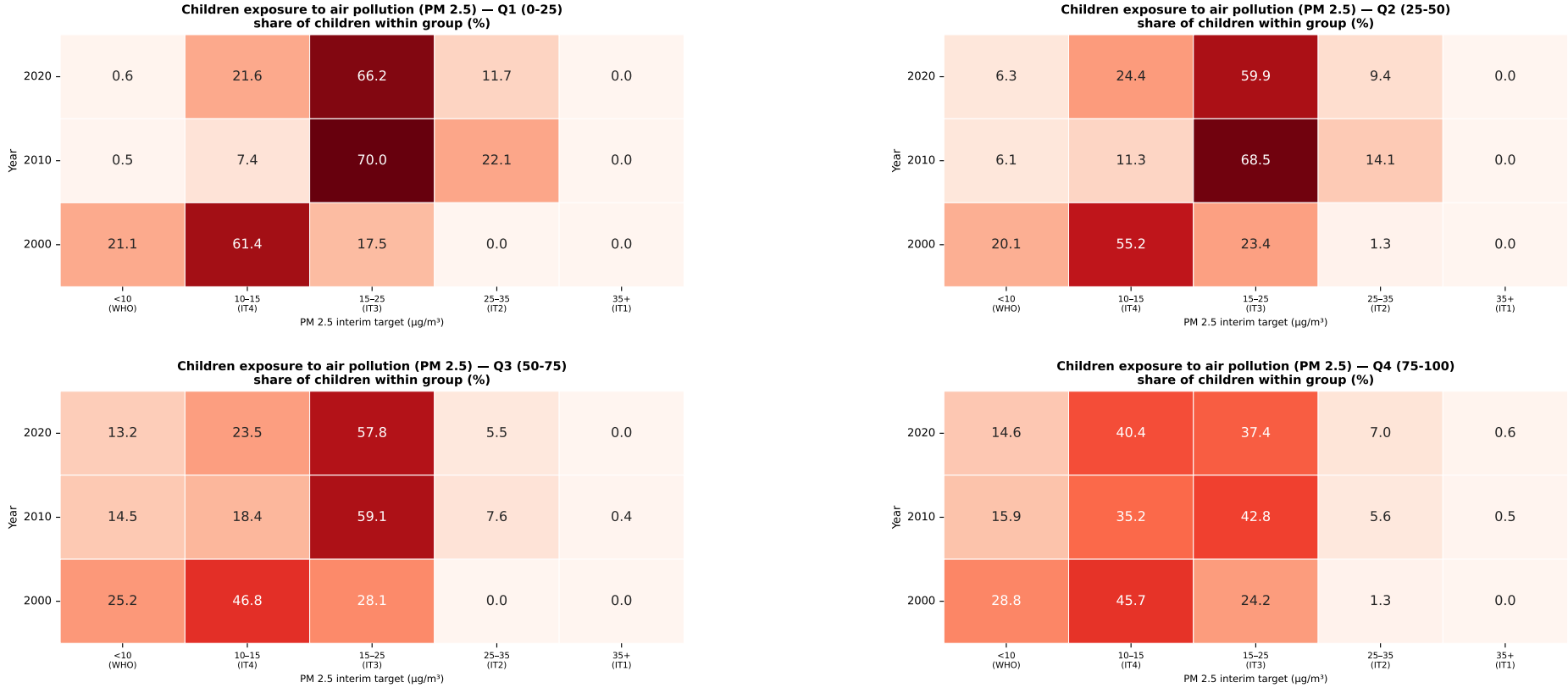
Notes: These heat maps show the number of children in millions exposed to different pollution threshold in 2000, 2010, and 2020 by income group.

Figure D.13: Children's air pollution exposure by HDI group



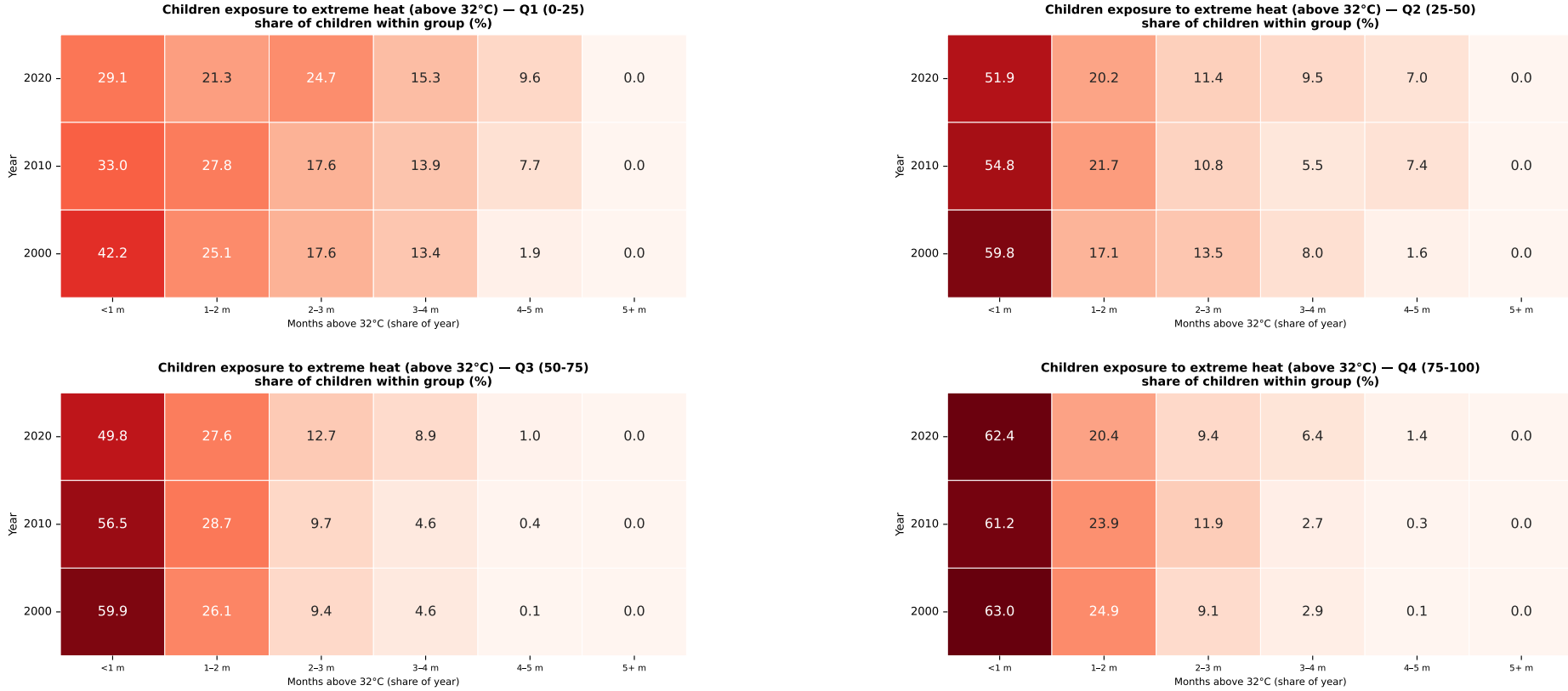
Notes: These heat maps show the number of children in millions exposed to different pollution thresholds in 2000, 2010, and 2020 by HDI group. HDI quartiles are based on 2020 subnational Human Development Index values, held constant across years (Q1 = lowest HDI, Q4 = highest).

Figure D.14: Children's air pollution exposure by accessibility group



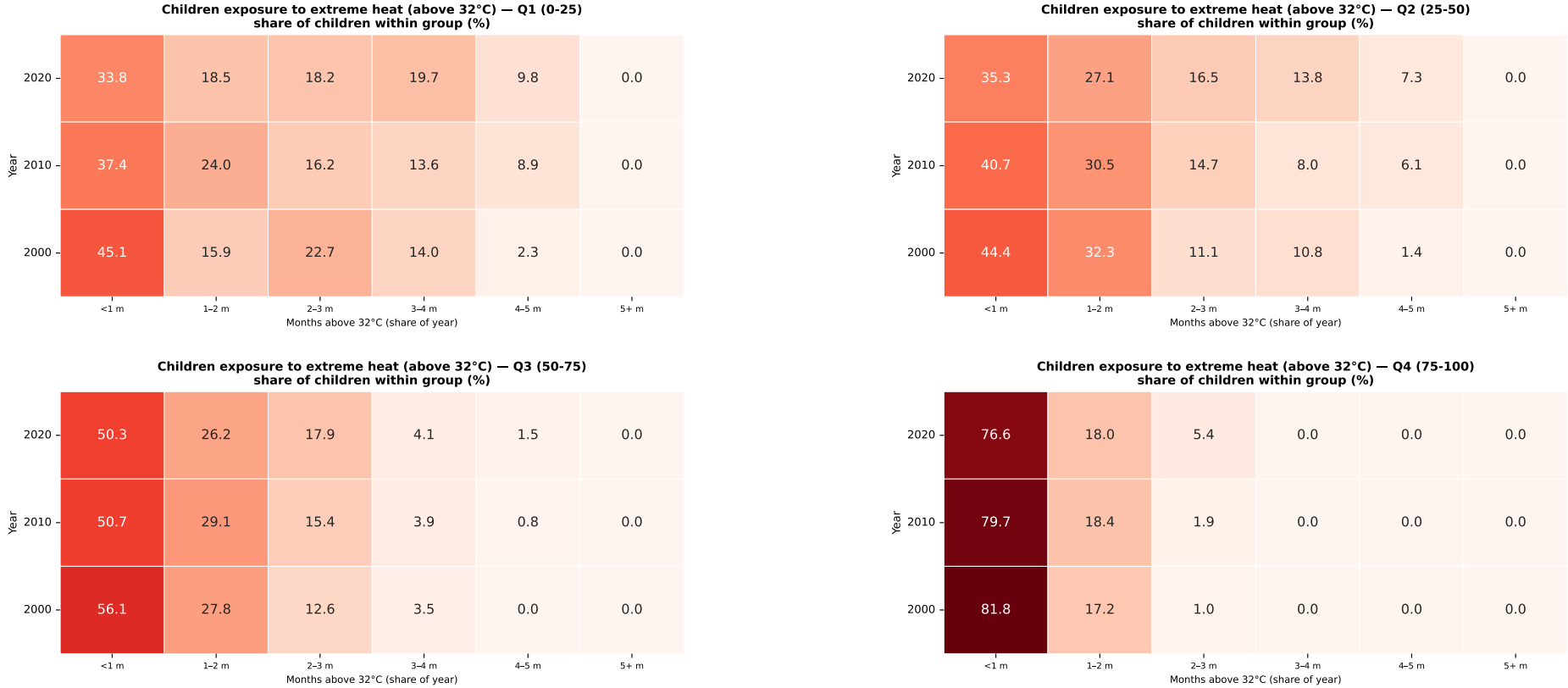
Notes: These heat maps show the number of children in millions exposed to different pollution threshold in 2000, 2010, and 2020 by accessibility group

Figure D.15: Children's heat exposure by income group



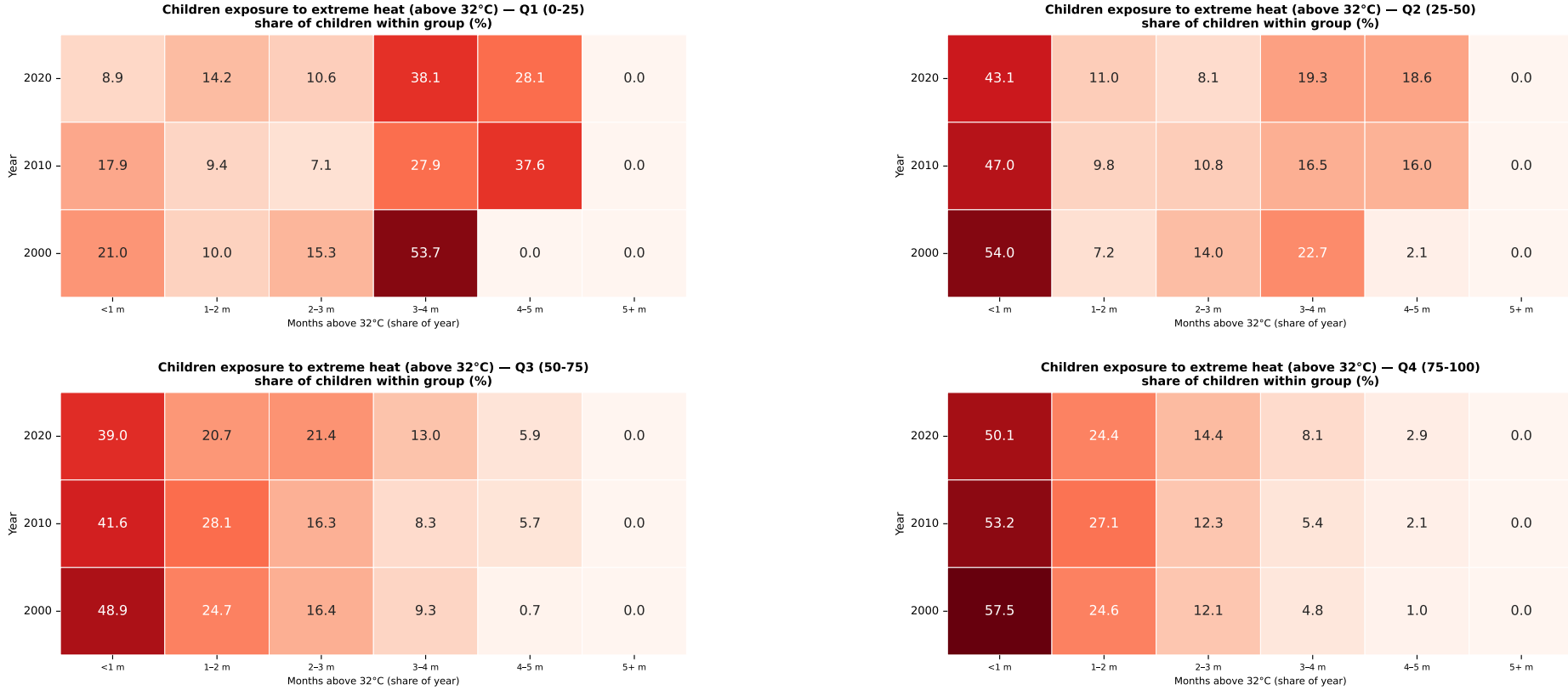
Notes: These heat maps show the number of children in millions exposed to different temperature threshold in 2000, 2010, and 2020 by income group.

Figure D.16: Children's heat exposure by HDI group



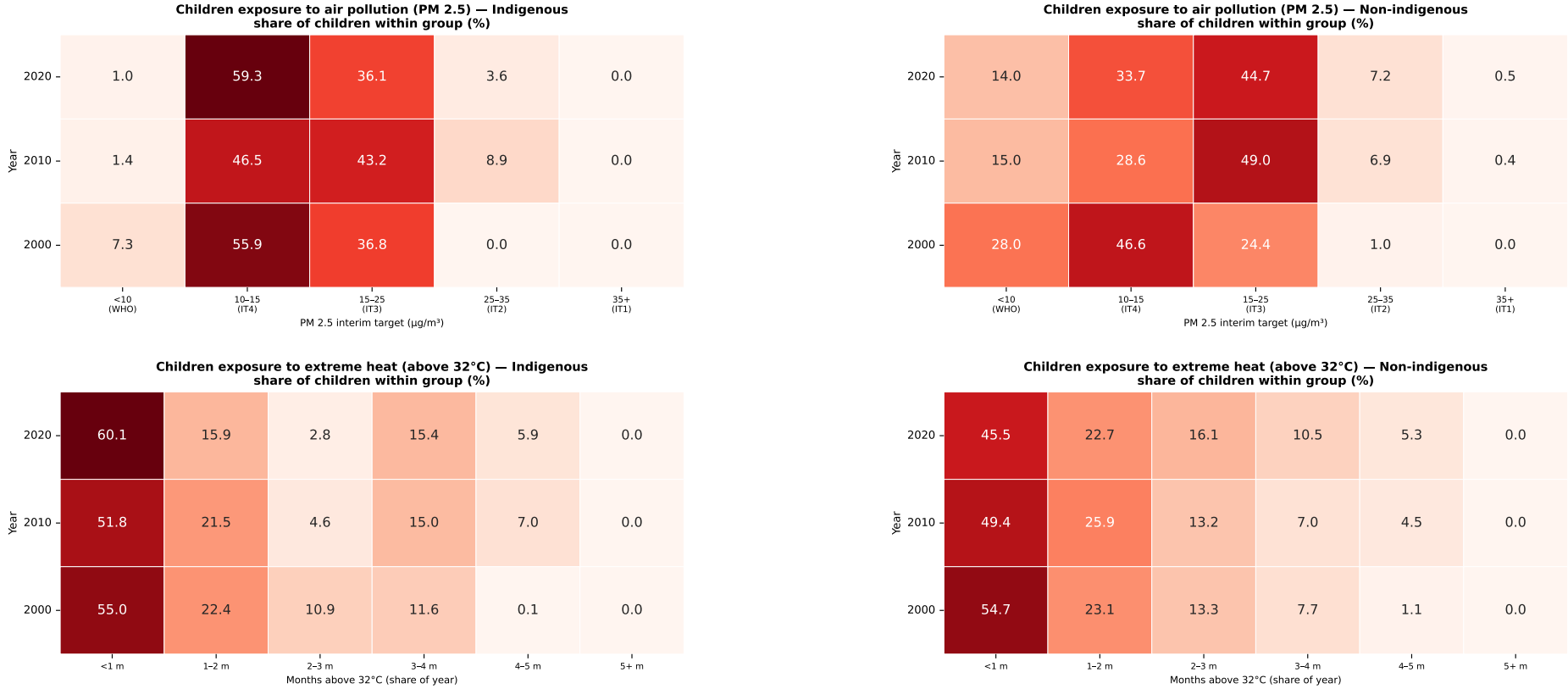
Notes: These heat maps show the number of children in millions exposed to different temperature thresholds in 2000, 2010, and 2020 by HDI group. HDI quartiles are based on 2020 subnational Human Development Index values, held constant across years (Q1 = lowest HDI, Q4 = highest).

Figure D.17: Children's heat exposure by accessibility group



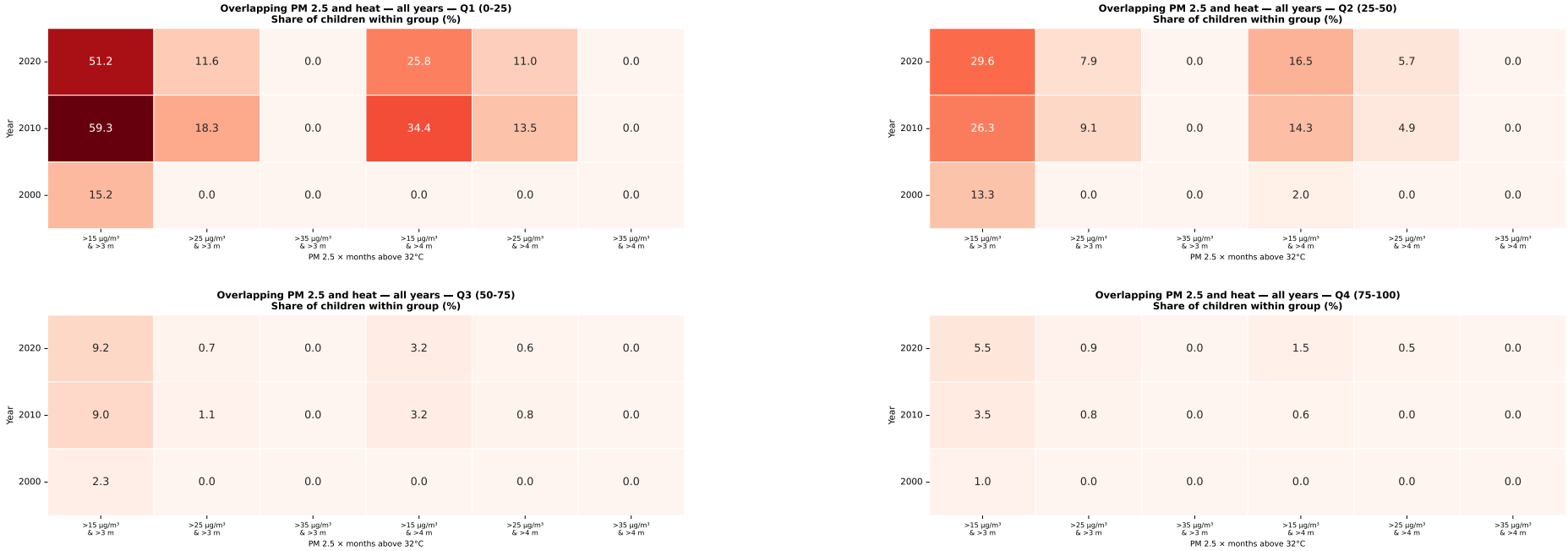
Notes: These heat maps show the number of children in millions exposed to different temperature threshold in 2000, 2010, and 2020 by accessibility group.

Figure D.18: Children's air pollution and heat exposure by land type



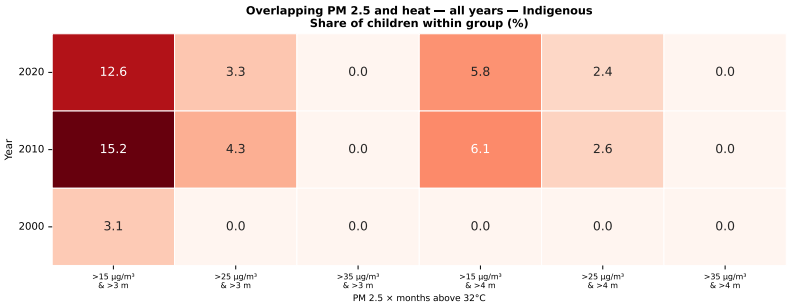
Notes: These heat maps show the number of children in millions exposed to different air pollution and temperature threshold in 2000, 2010, and 2020 by land type group.

Figure D.19: Overlapping air pollution and heat exposure by accessibility group

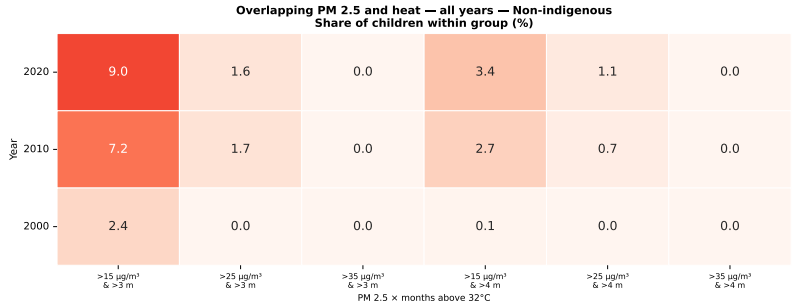


Notes: Share of children within each accessibility group exposed to combined $PM_{2.5}$ and heat thresholds, shown across all years (2000, 2010, 2020). Accessibility groups are defined by travel time to the nearest city, with Q1 the most remote and Q4 the most accessible quartile.

Figure D.20: Overlapping air pollution and heat exposure by land type



Indigenous



Non-indigenous

Notes: Share of children within each land type group exposed to combined $PM_{2.5}$ and heat thresholds, shown across all years (2000, 2010, 2020). Values reflect the percentage of children in indigenous and non-indigenous territories jointly above a given air pollution and temperature threshold.